

AI-Enhanced Platform for Personalized Hobby Improvement

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**A project report submitted in partial fulfilment of the
requirements for the award of Bachelor of Science (Hons) Software
Engineering with Honours**

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September 2024

DECLARATION

I hereby declare that this project report is based on my original work except for citations and quotations which have been duly acknowledged. I also declare that it has not been previously and concurrently submitted for any other degree or award at UTAR or other institutions.

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CHAPTER 1

INTRODUCTION

1.1 Background of Project

In the present generation, many platforms and applications can be used to foster people's hobbies and interests. However, few platforms offer users suggestions and recommendations, which are only offered in certain fields or interests. It is often limited to giving general information and materials and does not consider the individual user's choice, experience, and passion. This lack of personalization may result in low motivation and stasis and decrease the willingness to further engage in one's hobbies. The reason for having a personalized hobby improvement platform is quite evident. People require information and recommendations that will help them advance and accomplish their objectives in areas of concern to them. Customized solutions can greatly improve the user experience by offering useful and timely recommendations, which in turn will encourage further use of the service. Thus, in my project, I will create a hobby and interest enhancement platform using AI APIs that will help improve hobbies and interests. The platform will be based on AI technology and data analysis of user's preferences, skills, and interests. This way, users are provided with the most appropriate and specific information to help them enjoy their hobbies to the fullest.

1.2 Motivation

The motivation behind the creation of an AI-enhanced platform for improving hobbies is the absence of individual hobby enhancement. Some of the existing platforms do not offer relevant assistance and information to the potential hobbyists and therefore the intended beneficiaries are not able to develop their hobbies properly. This project shall seek to solve this problem by employing the OpenAI API and other information data resources to evaluate user's preferences, competencies, and interests. It can offer recommendations for hobby activities, tips for skill improvement, and content that improves user's hobbies. It enriches the hobby experience by giving detailed and functional breakdowns and recommendations, so they can enjoy their interests.

1.3 Problem Statement

1.3.1 Lack of user-friendly hobby management interface

Most platforms that are associated with hobbies suffer from interface clutter and poor navigation. This is annoying to the users and lowers their overall interaction with the application. This problem is not exclusive to hobby platforms – a typical pattern that can be observed in digital tools is the emphasis on looks over usability. People are often presented with interfaces that are hard to understand and that overwhelm them (Hotjar, 2022). Therefore, the management of hobby activities becomes inefficient. This makes it difficult for the users to monitor their progress and this affects their motivation, hence their satisfaction. Studies show the severity of this problem: 83% of users leave websites that have a poor interface design and 40% of them will never come back (Tseluyko, 2022). This goes to show that the issue of a good user interface is very crucial when it comes to any hobby management platform.

1.3.2 Limited Content and Resource Availability

Most hobby platforms currently in the market are limited in scope and mainly address particular hobby sectors, and most of the time, they rely on expert coaches for content dissemination. This leads to the absence of adequate quantity and quality of materials that the users require to enhance their competencies. Furthermore, the content that is available on such platforms is often old or too general and does not meet the needs of the individual users. Lack of organization and proper indexes aggravates the problem of finding the necessary information for users. This makes it difficult for self-motivated learners since the instruction is mainly by expert coaches. Also, the lack of recommendations for each user decreases the satisfaction level of the users since they are provided with recommendations that may not suit their preferences or their level of expertise. Increasing the variety and quality of content, organizing it properly, and using AI-based recommendation systems can dramatically improve the experience. The literature shows the importance of quality and easily available resources and personalised learning (Balderas-Solís et al., 2022).

1.3.3 Inadequate Personalized Recommendations for Hobby Improvement

Today's hobby platforms do not have a feature that allows them to suggest something that might be interesting to the particular user, their level of expertise, and preferences. The main issue is observed in the recommendation engine where the algorithms applied are either outdated or not sophisticated enough. These do not capture all the aspects of the user preferences and their actions in the particular context. Since the platform is not capable of recommending items based on the user's preferences, the user is presented with items that are not relevant to the user or the expertise of the user. This results in a negative experience for the user. If recommendations are not relevant to the user's needs, the user's progress and satisfaction levels are impacted negatively. On the other hand, the targeted recommendations can boost the level of user engagement to a significant degree. If recommendations are interesting and relevant, users will come back to the platform, which will increase customer loyalty (Banuba, 2023). Loyal customers also imply that the business will be able to retain its customers for a longer time, which is always a plus for the business (Banuba, 2023). Last but not least, the recommendations allow users to advance in the hobbies they have and achieve the set objectives.

1.4 Goal and Objectives

1. To design a user-friendly frontend interface and backend services for managing users' hobbies platform.
2. To integrate users' hobbies platform with various resources and discussion content for engaging platform's users to improve their hobbies.
3. To integrate Generative AI that analyze user preferences and hobby-related data, providing intelligent recommendations for improving their hobbies.

1.5 Scope

This project will create an AI-enhanced personalized hobby improvement platform. This project will be created by designing and implementing a clean and easy-to-use front end with the help of today's web application frameworks like Laravel, React, Angular, and Vue.js and many other robust back-end services like MySQL, PostgreSQL, MongoDB for hobby management and GUI. It will also need to be designed with a large content base to be able to offer thousands of materials, lessons, and resources that would be beneficial to the various levels of enthusiasts. Recommendations, improving the user experience, and providing suggestions for the improvement of their hobbies will be made with the help of OpenAI's API and AI recommendations. The efficiency, effectiveness and reliability of the system will be tested through conducting tests.

1.5.1 Primary User

This individual hobby enhancement platform is to be directed at Hobby Enthusiasts, beginners, experts, and users with broad interests to help them receive specific information.

1. **Hobby Enthusiasts:** People who are interested in their activities and desire to enhance their abilities and information in their fields.
2. **Beginners and Experts:** Provide for those who are beginners to the craft as well as those who are very experienced in it.
3. **Broad Interest Users:** People who have a general interest in different kinds of hobbies receive specific recommendations and information.

1.5.2 Platform Functionality

The platform features of this project include the following aspects, which are focused on the improvement of the user experience and ensuring efficient hobby management.

1.5.2.1 User Authentication

User authentication is to ensure that the users have a secure way to log in, log out and register for a new account for first-time users. Secure login and logout to enable the users to log in and out of their accounts safely is important in the security and ease of account access. It also incorporates a friendly and easy registration procedure for new users. Also, a password recovery system will be incorporated to assist users in securely retrieving their accounts.

1.5.2.2 User Profile Management

The user profile management is a feature that enables the user to modify personal information and hobby management anytime. These features enable the users to change their details and hobbies of interest at any time and also to add, modify and remove hobbies of interest. This feature makes it possible to provide a personalized recommendation to the users of the platform hence enhancing the user experience.

1.5.2.3 Hobby Recommendation System

The hobby recommendation system incorporates artificial intelligence algorithms (such as those from the OpenAI API) to evaluate the user's preferences and the data concerning hobbies. The platform will process the collected data of user preferences and information concerning hobbies using artificial intelligence. After that, the use of AI is applied to suggest intelligent and preferred activities, resources, and skills improvement.

1.5.2.4 Search and Discovery Function

The search and discovery features are important in order to allow the users to find hobbies and resources of their interest. There will be a strong search bar to search hobbies, articles, videos, and tutorials. Moreover, the discovery function will recommend new hobbies depending on the user's preferences and their history, and the recommended hobbies will be popular and hot in order to guarantee the user's activity.

1.5.2.5 Content Management Functions

The content management function is to develop the number of articles, videos, and tutorials about the various hobbies to assist users in finding hobby materials and improving their hobby level. It also allows the users to post, submit and share any content that they have that is related to the hobby and which other users can read and learn from.

1.5.2.6 Notifications and Alerts Function

Notification and alert functions are important in order to keep the users engaged and informed. The platform will send hobby activity alerts at certain times, inform users of new resources and changes, and enable users to specify the kind of notifications they receive and how often they receive them.

1.5.2.7 Scheduler Function

The scheduler function assists the users in the organization of their hobbies and related activities. The platform provides the users with the recommended daily, weekly, and monthly schedules that the users can follow. The use of calendar applications will also enable the synchronization of schedules and reminders while the features of tracking progress will enable the users to track their progress.

1.5.2.8 Admin Dashboard

The Admin Dashboard is a tool that has been created to be used by administrators to control the users and their activities on the platform. The Admin Dashboard can view the content created by the users. According to the results of the review, the administrators allow or reject the content created by users to maintain the quality and relevance. The Admin Dashboard gives information about the utilization of the platform, the users, and the content.

1.5.3 Out of Scope

Some of the areas of the project that are excluded are the following: The project will not involve the creation of a mobile application; the project will be a web-based one. At the start of the platform, there will be no support for multiple languages, and the platform will only be in English. Connection with other

social networks for user identification or content posting/retrieval will not be supported. Real-time speech recognition or any other sophisticated natural language processing capabilities are also not considered, beyond recommendation and personalization. Also, the platform lacks the integration of e-commerce features.

1.6 Problem Solution

1.6.1 User-Friendly Hobby Management Interface

The problem that the proposed solution attempting to address is the fact that most hobby management platform interfaces are difficult to use. The platform will not be filled with features that are not needed and that will just complicate the users and distract them from their hobbies. Among the design principles that will be applied are simplicity of the visual organization, usability and responsivity. With the help of feedback and usability testing, the interface of the platform will be adjusted to the needs and requirements of the user. This approach is believed to enhance the level of user participation and satisfaction thereby decreasing the level of abandonment and enhancing the level of loyalty to the platform.

1.6.2 Improving Hobby Platform Content and Accessibility

To eliminate the problem of a limited amount of content and the lack of materials, it is necessary to expand the list of hobbies and the complexity of the content offered. Using AI API, one can offer the users content that will be relevant to them and meet their level of knowledge. The changes in the organization and the search will help the users to find the content that they want easily. Every user can upload their methods, experiences, and tips regarding the improvement of their hobbies. This will help in making sure that the content is posted as frequently as possible so that it can be as current as possible. Through this approach the platform will be more interactive and helpful to the users hence improving on their experience and hence loyalty.

1.6.3 Personalized Recommendations for Hobby Improvement

In order to eliminate the weaknesses associated with the use of personalized recommendations for the improvement of hobbies, the proposed solution involves the use of AI algorithms that will analyze the user's preferences, skills and interests. The platform will be using OpenAI API to create recommendation systems that will suggest activities, resources, and improvements to the skills of the user. These recommendations will be updated through machine learning and will be adjusted based on user's behaviour and their feedback. Thus, the platform will improve the user's experience and their progress in hobbies by the feedback that will be provided to them. This is a technique that seeks to ensure that the users are in a position to gain what they want from the site hence making them to be loyal to the site.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

In chapter 2, I will perform a literature review to create a reference and background for creating an AI-enhanced platform for personalized hobby improvement. AI-enhanced platform are digital environments that employ artificial intelligence (AI) to assist users in improving their hobbies. Hobby enhancement for individual users will assist in the identification of hobbies that suit the user's interests, abilities, and choices. This chapter will define gaps and desires in the market based on the analysis of existing platforms focused on hobbies. Furthermore, this chapter will propose the development methods for AI integrated platforms and the various AI platforms and software development tools that are appropriate to be used in this project such as backend, frontend, and database. In conclusion, this research seeks to advocate for the most effective strategy in creating the most effective and user-oriented AI-based platform for enhancing hobbies.

2.2 Existing Hobby Enhanced Platforms

2.2.1 PLAY.FUN App



Figure 2.2.1 PLAY.FUN Logo

The PLAY.FUN app is a paid service that complements the sports by providing classes and other related activities. PLAY.FUN targets its target audience to

create an active community and offers a list of available sports: tennis, beach volleyball, basketball, yoga, and other sports suitable for beginners and professional players (PLAY.FUN, 2019).

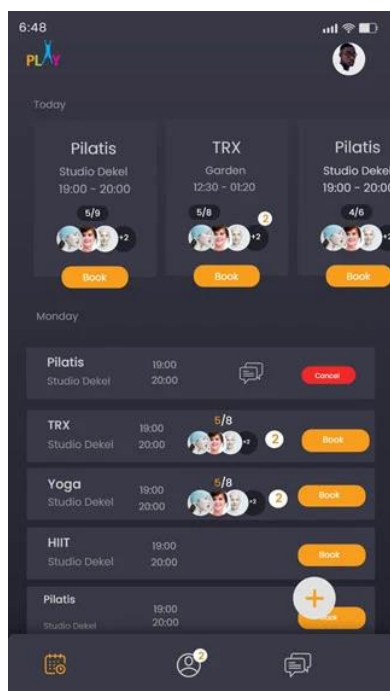


Figure 2.2.2 Main Page of PLAY.FUN App

One of the most significant advantages of PLAY.FUN is that it has very flexible membership options. These plans start from Lite and go up to Gold and the monthly cost is \$50 to \$200. The premium plans provide more specific training and VIP features, which will enable the users to customize their fitness experience (PLAY. FUN, 2019). To attract new users, PLAY.FUN offers free trial classes to members so that potential members can experience different activities before subscribing (PLAY.FUN, 2019).

Coaching by professionals is one of the essential elements of PLAY. FUN. The classes are conducted by professional coaches and give the users recommendations on how to enhance their abilities (PLAY. FUN, 2019). The availability of many classes in the app makes the user try out new sports and hence improve on their skills. Also, the social component of PLAY. FUN enables users to communicate with other like-minded individuals, exchange

information, train, and encourage one another to become better (PLAY. FUN, 2019).

In summary, PLAY. FUN is an application that offers professional training, variety of exercises, and a friendly community to improve sports and fitness abilities with fun. The app offers open membership and free trial classes so that the user can select the right program that will keep him or her active and engaged in the fitness program.

2.2.2 Blinkist

Blinkist is an application that provides the main ideas of more than 6,500 non-fiction books and podcasts in the form of 15-minute reads or listens. Blinkist is perfect for people with little time to learn new things and is available in such categories as business, self-development, psychology, and relationships. Thus, Blinkist provides users with new viewpoints and improves their perception of the world by offering brief and practical summaries (Blinkist, 2013).

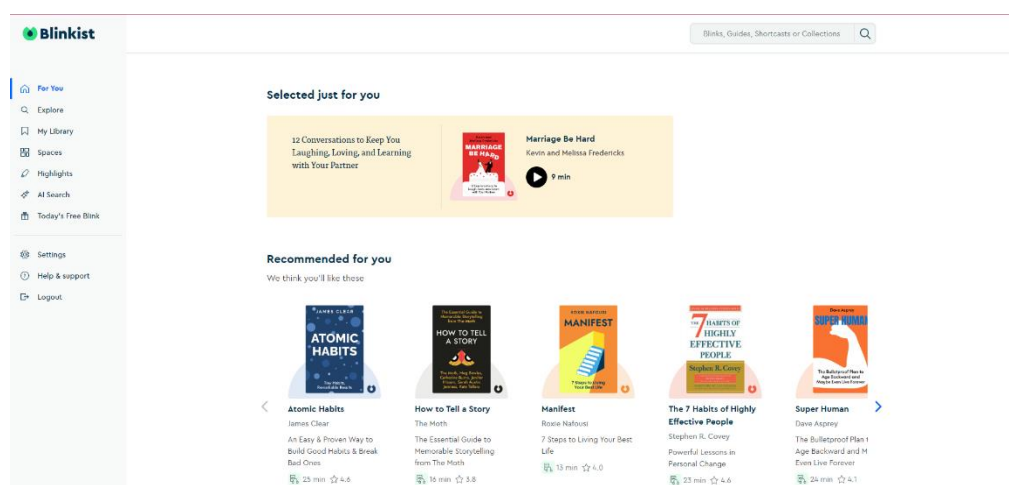


Figure 2.2.3 Main page of Blinkist

To enhance the user's hobby levels, Blinkist has the following aspects. First, it gives users a chance to read articles and books from various fields, which are chosen by professionals, so people can discover new topics they are interested in or develop their existing hobbies. The Shortcasts that the app provides as the brief of the main points of popular podcasts also provide more

tips and ideas for the users who need to develop themselves and learn more (Blinkist, 2013).

Blinkist's summaries are written by a team of professionals based on the principles of neuro-psychology, so the information is easy to digest. This approach assists the users in gaining an understanding of the fundamental concepts of a book or podcast so that they can apply such knowledge to their hobbies. In addition, the app's audio option enables the users to learn while doing other activities like listening to a brief on cooking methods while preparing food or a podcast on photography while editing pictures (Blinkist, 2013).

New titles and recommendations are added to the application depending on the current trends, so users always have an opportunity to expand their knowledge and develop their skills and interests. Blinkist offers specific suggestions and provides a vast amount of information to keep the users motivated and knowledgeable about their goals (Blinkist, 2013).

2.2.3 Yousician

Yousician is an application aimed to teach and improve the skills of playing musical instruments such as guitar, piano, ukulele, bass, and singing. With the help of the lessons provided on the site and the feedback given in real-time, the learning process is entertaining and productive. The app listens to your playing, gives feedback on the accuracy and timing, and lets you know when you are correctly playing the right notes and chords. It has thousands of lessons and exercises for players of all levels, from novice to professional, and for all styles and songs (Yousician, 2014).

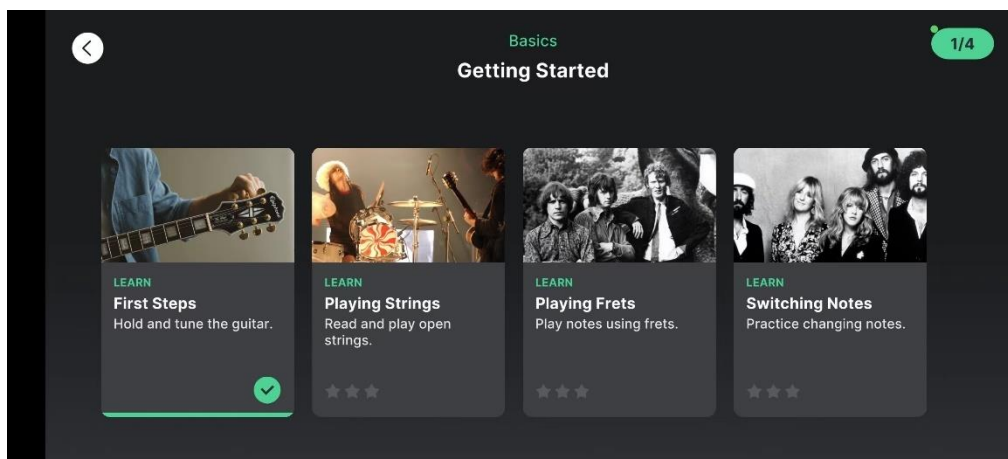


Figure 2.2.4 Main Page of Yousician App

To enhance the user's hobby levels, Yousician provides lesson plans and exercises that are selected by the experts, allowing the users to develop and polish their skills step by step. The users can then be taken through a series of lessons that will help them move from the basic level to the advanced level; for instance, chord, rhythm, and music theory for guitar or hand coordination for piano. The feedback provided in the app is real-time, which means that the user can correct the mistake as soon as possible thus increasing the learning effectiveness. In addition, the provision of a large number of songs and exercises can help users play their favourite tracks to maintain motivation and interest. Since it can be used on multiple devices, Yousician can be practised at any time, thus becoming a part of the daily schedule effortlessly (Yousician, 2014).

Thus, Yousician provides users with an effective and engaging tool for improving their musical hobbies and skills.

Table 2.1 Comparison between Software Development Methodology

Platform	Advantages	Disadvantages
PLAY.FUN	- Organized class reservations and membership management. - Numerous sports facilities. - Offers membership plans and expert coaching.	- Lacks features for hobby management. - Focuses solely on sports. - No AI-driven recommendations.

Blinkist	- Easy access to summaries and Shortcasts. - Extensive collection of book summaries and podcasts. - Personalized content suggestions.	- Lacks tools for hobby progress monitoring. - Summaries may lack detail for specific hobbies. - Personalization limited to content recommendations.
Yousician	- Interactive lessons with real-time feedback. - Extensive lessons and exercises for various instruments. - Personalized lesson plans.	- Limited to music-focused features. - No broader hobby management. - Personalization is instrument-specific.

2.3 AI-Enhanced Platform

2.3.1 ChatGPT

ChatGPT is an AI-created chatbot service launched and managed by OpenAI based on the GPT-3 and now GPT-4 model. At first, as seen in the performance of the work called 'ChatGPT,' the model integrated with GPT-3 did a brilliant job with the tasks related to natural language understanding and generation, including text writing, translating, and summarizing. The following enhancement was launched on March 14, 2023, GPT-4: Able to generally be steered, be multimodal, be more creative, and write code (Ticzon, 2024).



Figure 2.3.1 Logo for ChatGPT

ChatGPT has a simple interface reminiscent of instant messaging apps, making it accessible even to users without coding experience. Regarding the pinnacle abilities it does very well in the areas of speed, accuracy and fluency and these are highly valued in customer relations, teaching and entertainment. It is quite good for conversational interactions but may have some shortcomings in code-related issues in comparison with other similar tools (Ticzon, 2024).

As for integration and scalability, ChatGPT can be easily integrated with applications like messaging platforms, social media, and websites; it is capable of managing multiple interactions at once. It uses a freemium model that offers a basic version for individual use and a pro version for business use if the usage is much higher (Ticzon, 2024).

In general, the advantage of ChatGPT is that it is conversational and simple to use, making it versatile but it may not have as many features as some of the other models out there (Ticzon, 2024).

2.3.2 Gemini

Gemini is an artificial intelligence-based content optimization software that was created by MarketMuse to improve content on websites using natural language

processing and machine learning. Derived from the natural language understanding (NLU) research, Gemini provides the list of the most effective web pages and examines their layout to determine the strengths, weaknesses, opportunities, and threats. While ChatGPT is more powerful in writing content and creativity, Gemini is designed to analyze the content and offer improvements, including the use of keywords and topics for better search engine results (Ticzon, 2024).

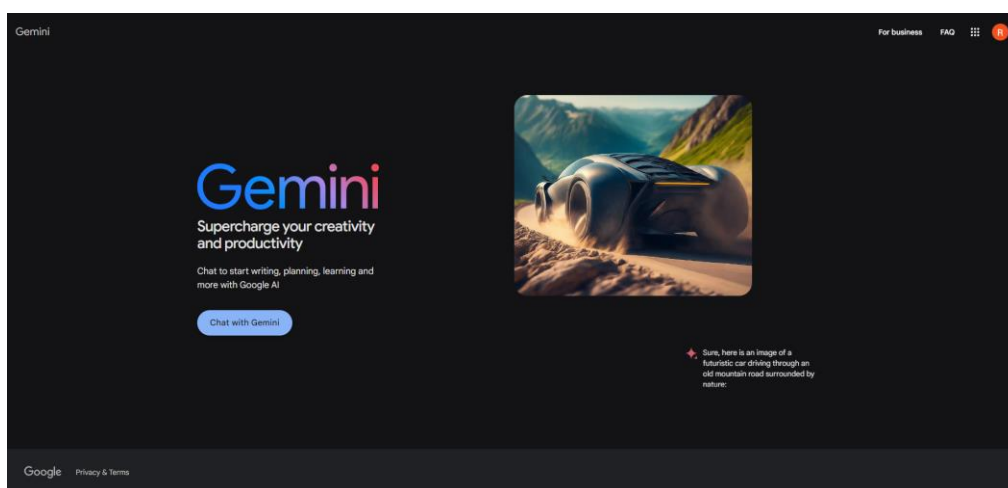


Figure 2.3.2 Main Page of Gemini

The Gemini client has a simple layout and can be used by both novices and experts; however, some users have experienced glitches and slow processing at times. It has features like sentiment analysis, keywords extraction and content enhancement and therefore is beneficial in SEO. However, it is not as fluent in the conversation as ChatGPT is, as observed by Ticzon in 2024.

Compatibility with other CMS and social networks, as well as available scale-up solutions, enables Gemini to deal with large content generation tasks. Freemium is the business model, but to use it one has to submit an application, which can take time. However, its pricing model and interface might be a drawback for some users despite it has sophisticated NLP capabilities and business-related features (Ticzon, 2024).

2.3.3 Copilot

Copilot is an AI code suggestion tool developed by OpenAI in collaboration with GitHub. Launched in June this year, Copilot uses billions of lines of code from GitHub to provide relevant and contextual code suggestions in real time. This functionality makes it more efficient than the conventional auto-complete tools and utilities. Among these, the primary and one of the most useful is the capability to create whole functions or classes from the input of the user, which greatly helps to save the time of a developer, especially if it is necessary to write a large amount of similar code or so-called ‘boilerplate code’ (Ticzon, 2024).

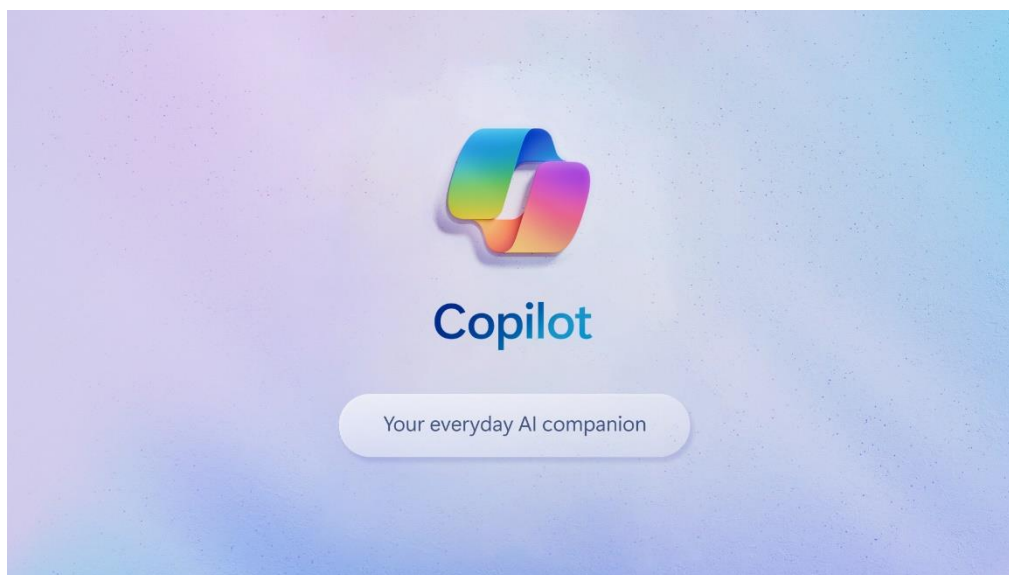


Figure 2.3.3 Logo of Copilot

Available as a plugin to be installed in such platforms as Visual Studio Code or GitHub, Copilot operates in the background, providing suggested snippets as the developer writes code. This integration is intended to increase the speed and reliability of coding work. However, due to Copilot’s emphasis on code-specific tasks, it will not be as good at basic conversation as ChatGPT and Gemini (Ticzon, 2024).

Still, Copilot is in beta and is only accessible to a limited number of users, but it has demonstrated potential with its machine learning-based contextual features. It is expected to be released officially with a freemium model like its predecessor ChatGPT and its counterpart Gemini. To sum it up,

Copilot's advantages include its versatility and the ability to improve SEO approaches with the effective application of such measures as structured data markup and website loading speed (Ticzon, 2024).

Table 2. 2 Comparison between AI-Enhanced Platform

	Advantages	Disadvantages
ChatGPT	- Generates human-like text; versatile. - User-friendly with fast response time. - Scalable; integrates with various platforms. - Freemium model with easy access.	- Possible inaccuracies. - Limited business features. - Less effective for coding.
Gemini	- Advanced NLP; ideal for content optimization. - Clean, easy-to-navigate interface. - Fast, accurate responses. - Integrates with CMS and social media. - Freemium model with application access.	- Potential bugs; complex interface. - Subscription fees. - Less natural conversation than ChatGPT.
Copilot	- Accurate code completion. - Seamlessly integrates with code editors. - Efficient for coding tasks. - Scalable for coding projects. - Freemium model in beta.	- Limited to coding; not conversational. - Requires significant data for training. - Not suitable for casual conversation.

2.4 Software Development Methodology

2.4.1 Agile Software Development

Agile software development is a model of software development that is based on the iterative and incremental model and that is characterized by flexibility, collaboration and customer feedback. This methodology is different from other methodologies like the waterfall methodology in that it focuses on delivering small and easily manageable software increments. Another concept in Agile is customer interaction, and this is because the end user is involved in the development process and hence it is easier for teams to handle changes in requirements. The main benefits of Agile are the improved focus on the users' needs, the early detection of issues, and the ongoing provision of a functional application (Brush & Silverthorne, 2022).

The concept of delivering individual parts of the software that Agile promotes means that organizations have to change the culture within the teams and accept the fact that collaboration and flexibility are essential. It helps teams to work in an environment characterized by change while at the same time guaranteeing that business value is created and delivered. Agile improves the productivity of organizations because the teams are well organized and all the people understand their responsibilities. Moreover, testing that occurs during the development stage helps to achieve high-quality of products make changes and find potential issues (Brush & Silverthorne, 2022).

Some of the methodologies that are part of Agile are Crystal, Scrum, Feature-Driven Development, Extreme Programming and many more. All of these approaches mean making the software in increments with mini-increments of new functionality. This is because Agile's iterative approach means that the benefits of the software can be realised early and there are frequent small improvements; this makes efficiency better as it is easier for teams to identify and fix defects (McGuire, 2024).

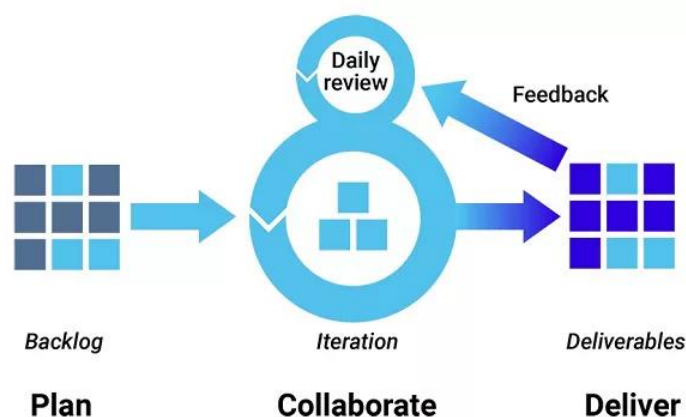


Figure 2.4.1 Agile development methodology

However, it is not devoid of some disadvantages which are as follows: It is very much dependent on real-time communication and sometimes new users are not given crucial documents. In addition, Agile requires a lot of time from users and can be tiresome since the developers have to complete all the features to the satisfaction of the user within an iteration. The adoption of agile can be challenging for organizations, especially where the organization is large or has always worked with the waterfall approach; this is why the combination of the two is sometimes the best approach (McGuire, 2024).

2.4.2 Waterfall Software Development

Waterfall software development is a traditional, linear methodology characterized by its sequential phases. There are five phases that it consists of, these are the requirement phase, design phase, implementation phase, verification phase, and maintenance phase. All the phases must be done and there is no way to go back and modify something in the project or the direction of the project. This rigid structure makes the Waterfall model implement and control, especially for projects with well-defined goals and fixed specifications (McGuire, 2024).

The Waterfall Method



Figure 2.4.2 Waterfall Method

The Waterfall method starts with the collection and writing down of user requirements that act as the basis for the design phase. This design phase also involves developing a system and software architecture. The final step is the implementation phase where coding and building of the software takes place after the design phase is over. Lastly, the testing phase ensures that the software developed is right in line with the requirements set and is not flawed in any way. The last is deployment which is the process of putting the software into use and maintenance which is the process of supporting the software after it has been deployed to the users (Adobe Communications Team, 2022).

The Waterfall methodology is also very structured, and this makes it easy to define project scope and timelines, thus easy to manage. This predictability is especially helpful for junior project managers and their teams, or for projects where the team's composition is often changing. This is because the specific planning and documentation at the beginning of the project enable the estimation of time and the setting of release dates (Adobe Communications Team, 2022).

But the Waterfall model also has some serious disadvantages. This means that the structure is rigid and the controls are tight, which leads to slow and expensive project delivery. While working on a project, there are instances

when some of the parameters change in the middle of the process, and this is not easily possible in Waterfall because of its rigidity compared to Agile which is iterative. Such restrictions compel teams to look for other more flexible and effective development paradigms, which are discussed below (McGuire, 2024).

Nevertheless, it is still possible to consider the Waterfall methodology as a valid approach to projects that have clear and well-stipulated requirements and a linear workflow. Due to its focus on detailed planning and step-by-step implementation, it can be applied to some types of engineering and industries.

2.4.3 Rapid Software Development

Rapid Application Development is an iterative method of developing software, which is not as strictly defined in the planning stage as other models. Originating in the 1980s, RAD allows for continuous evolution based on user requirements, distinguishing it from the rigid, sequential nature of the waterfall model (Kissflow, 2024). The RAD process is comprised of four phases: These are the requirement planning, the user design, construction and the cutover. The user design and construction phases are iterative, repeating until the product meets user requirements (McGuire, 2024).

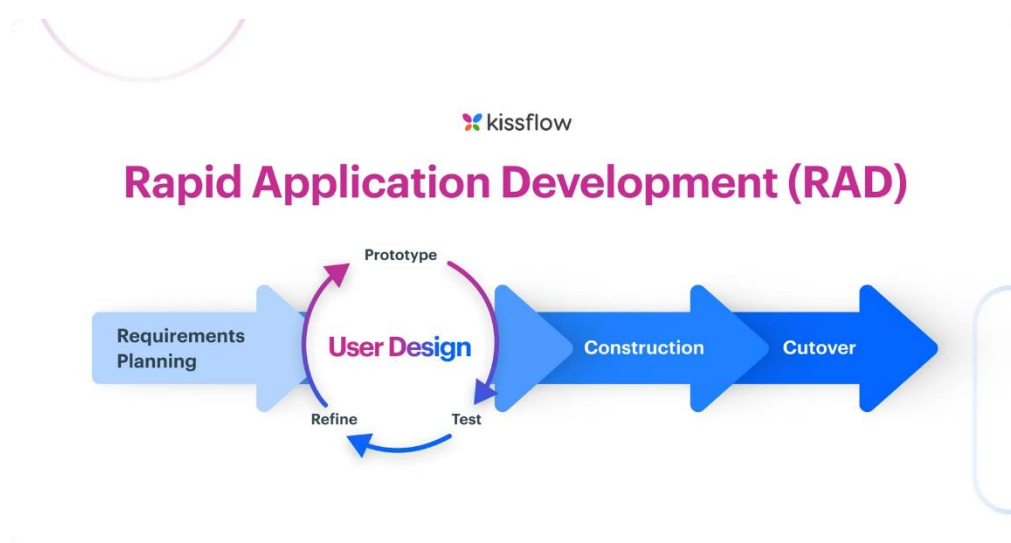


Figure 2.4.3 Rapid Software Development

A significant advantage of RAD is its ability to produce high-quality systems with lower investment costs. By enabling developers to swiftly adjust

to shifting requirements, RAD supports the development of prototypes that can be tested and refined without starting from scratch. This results in software that aligns closely with end-user needs and provides immediate value (Kissflow, 2024). The approach is most useful where there are clear business goals and the users are not complex, making it good for small to medium, high visibility projects that need to be completed within a short time (McGuire, 2024).

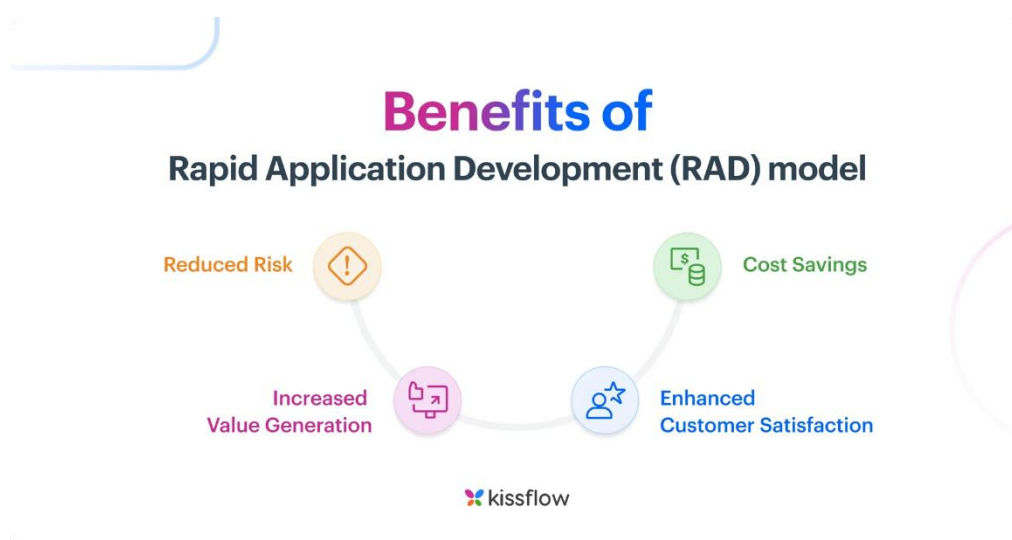


Figure 2.4.4 Benefits of Rapid Software Development

The advantages of RAD are time-to-market, cost advantage, more value creation, risk minimization, and improved customer satisfaction. The iterative development and prototyping allow for quick delivery of usable software parts, giving businesses a competitive edge. Continuous user feedback helps identify and rectify errors early, reducing extensive debugging costs and accelerating revenue generation. This is because through the use of RAD, users are involved from the development phase to the final product hence increasing the chances of the final product being satisfactory to the users hence increasing customer satisfaction and loyalty (Kissflow, 2024).

Nevertheless, RAD also has its disadvantages. It needs a stable team of highly skilled developers and knowledgeable users who can give feedback and approval on the spot. Organizations lacking these resources may struggle to benefit from the RAD approach. Additionally, the condensed development

timeline demands deep knowledge of the application area, which can be challenging for some teams to achieve (McGuire, 2024).

2.4.4 DevOps Software Development

DevOps is a software development model that merges software development (Dev) and IT operations (Ops) to reduce the development lifecycle and develop high-quality software. It integrates best practices that promote cross-functionality between development, quality assurance, and operations functions while promoting organizational culture. DevOps is based on CI/CD, which means that integration and deployment are constant; the use of automation for repetitive tasks; and collaboration in teams to enhance productivity and the quality of the software (GitLab, 2023).

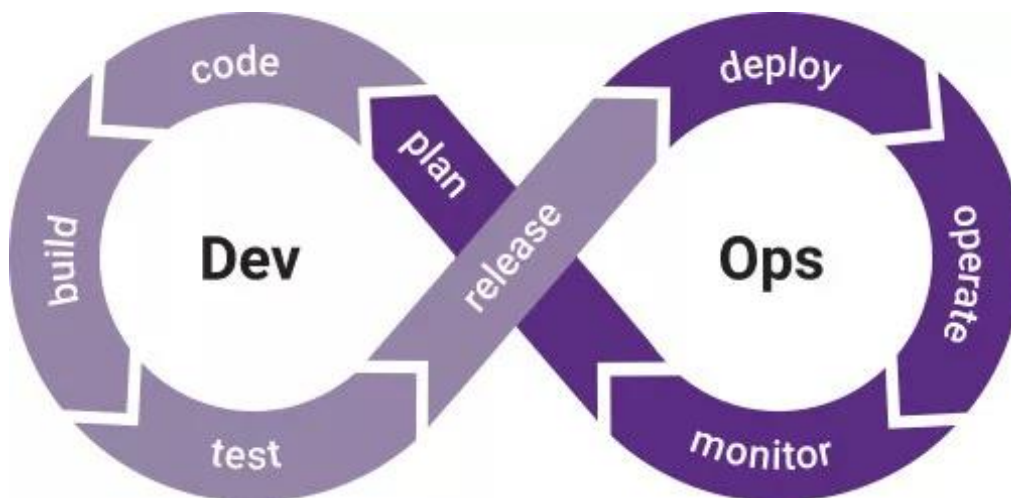


Figure 2.4.5 DevOps Development

The methodology is designed to increase time to market, reduce the rate of new product failures and reduce disruption while increasing reliability. DevOps is the set of practices that involve automation of the software development process, testing, builds, releases and deployment of the development environment. This automation helps to minimize human intervention and also helps in faster delivery (McGuire, 2024). DevOps is a set of practices that help to build, test, and release software with accountability, collaboration, and speed (GitLab, 2023).

DevOps is also characterized by a culture of constant improvement and the reduction of waste. The best DevOps teams always look for opportunities to optimize, for instance, by trying to minimize the time taken in a release or mean-time-to-recovery. Also, DevOps is centred on user focus by providing short feedback, which guarantees that user needs are met to the maximum (GitLab, 2023).

However, like any concept, DevOps has its disadvantages. It is not always good to be updating the system continuously and some industries have set rules that before going to the operational phase, certain tests have to be done. Also, if one department uses a different environment from another, problems can be introduced into the production environment unnoticed. Some of the quality attributes also include interaction with human beings which may slow down the pipeline of delivery (McGuire, 2024).

In general, DevOps enhances the time to market, satisfaction of customer, quality of product, and employees; performance. It also makes it easier to work at all phases of the development process since development and operations teams are combined, thus increasing the effectiveness and efficiency of application development and deployment.

Table 2.3 Comparison between Software Development Methodology

Software Development Methodology	Advantages	Disadvantages
Agile	- Responsive to user needs. - Early issue detection. - Flexibility to adapt to changes. - Continuous delivery.	- Time-consuming for users. - Heavy reliance on real-time communication. - Difficult for teams used to Waterfall.

Waterfall	- Easy to manage with clear scope and timelines. - Suitable for stable, well-defined projects. - Good for less experienced teams.	- Rigid structure; slow and costly. - Hard to adapt to changes. - High risk if initial requirements are flawed.
Rapid (RAD)	- Faster time-to-market with iterative prototyping. - Cost savings from early error detection. - Frequent releases increase value. - Reduced risk with incremental development.	- Requires skilled developers and strong user involvement. - Not ideal for complex projects
DevOps	- Faster time-to-market with CI/CD automation. - Improved quality and collaboration. - Continuous improvement.	- Some customers may resist continuous updates. - Regulatory constraints can slow deployment. - Different environments may cause undetected production issues.

2.5 Software Development Tools

2.5.1 Front-end Frameworks

2.5.1.1 React

React is a JavaScript library created by Facebook to building user interfaces, especially for single page applications. Introduced to the public in May 2013, React was developed by Jordan Walke and has since been considered one of the most important frameworks in front-end development (Javatpoint, 2013).



Figure 2.5.1 Logo of React

React is famous for its component-based approach which makes the process of building and maintaining the user interfaces easier. In React, the UI is made up of components that are reusable and each of the components is supposed to render a part of the interface. These components can be nested inside the other, which gives the developers the possibility to create complex applications from basic components which are easy to manage.

Another important aspect of React is the usage of the so-called virtual DOM or Document Object Model. Unlike other frameworks where the whole DOM is refreshed each time an update is made, React's virtual DOM updates only the affected nodes. This leads to enhanced performance and efficient rendering of user interfaces (Javatpoint, 2013).

React works mainly in the view component of an application and follows the MVC model. Although React is responsible for the ‘V’ layer, state and architecture are usually handled by libraries like Redux or Flux. This division of responsibilities enables React to concentrate on constructing the user interface while other libraries manage the state and application.

The library also has flexibility and community support which are other added advantages to the library. This makes React to be declarative in that developers describe how the UI should be and React will ensure that it renders it most efficiently. This makes React a tool of choice for modern web development, especially for the development of dynamic and interactive user interfaces (Javatpoint, 2013).

2.5.1.2 Angular

Angular is a comprehensive platform and framework developed by Google for develop single page applications (SPAs) using HTML and TypeScript. Angular, written in TypeScript, leverages a robust set of libraries to handle core and optional functionality, enabling developers to build dynamic web applications efficiently (Angular, 2023).



Figure 2.5.2 Logo of Angular

The core of Angular’s architecture is the components which are the basic elements for defining the interface. Each component controls a certain

view, which is a set of screen items that can be changed depending on the application's logic and data. Components are designed to be modular and reusable, which enhances code efficiency and maintainability (Angular, 2023).

Angular employs a robust feature called two-way data binding that binds the data model with the view in real-time. This means that changes in the data automatically reflect in the view and vice versa, simplifying the process of keeping the user interface and data in sync. Additionally, Angular employs dependency injection (DI) to manage the injection of services into components. Services provide background functionality, such as data retrieval, which is not directly related to the view but essential for application logic. This modular approach allows for more organized and maintainable code (Angular, 2023).

The framework also includes Angular directives and binding markup, which are used within templates to dynamically modify the HTML before it is rendered. This capability, along with Angular's Router service, supports complex navigation and routing within applications, facilitating sophisticated in-browser navigation (Angular, 2023).

Angular has a rich set of features that include a rich set of tools and libraries that can be used to build modern web applications. Some of the features include two way data binding, dependency injection, and hierarchical view management making it ideal for developers who want to develop efficient web applications.

2.5.1.3 Svelte

Svelte is a new way of constructing user interfaces that differs from the majority of other frameworks. Compared to such frameworks as React and Angular, which perform a significant amount of work during runtime in the browser, Svelte operates at build time and outputs optimized plain JavaScript. This pre-compilation process leads to smaller application bundles and better performance, which is especially efficient for web applications (Green, Chrisdavidmills & Opensas, 2024).



Figure 2.5.3 Logo of Svelte

Svelte does not use a virtual DOM, which is a feature commonly present in other frameworks. Rather, it gathers components into highly optimized, imperative code that directly manipulates the Document Object Model. This results in increased speed and efficiency since there is no need for a virtual DOM layer to monitor changes and update the interface.

The framework is designed to be easy to use and is built on top of the most basic web technologies: HTML, CSS, and JavaScript. It builds upon these technologies with little extra technologies, which means that the developers who already have some web development experience can easily learn and apply this framework effectively (Green, Chrisdavidmills & Opensas, 2024). This also leads to less boilerplate code and a more efficient process of development of the application.

Nevertheless, Svelte is significantly younger than the other frameworks listed above and, therefore, is not as developed. Its ecosystem is not as developed as the one of WordPress, which has more tools, plugins, and resources. As a result, the job openings may be limited and the challenges are likely to be greater for developers who are switching from other frameworks. Nevertheless, Svelte is rather suitable for building applications for low-power devices, highly interactive pages, and complex visualizations because of its minimal and optimized code output.

SvelteKit extends Svelte’s features with more functionalities such as filesystem-based routing, and server-side, and offline capabilities for modern web applications (Green, Chrisdavidmills & Opensas, 2024).

Table 2.4 Comparison between Front-end Framework

Front-end Framework	Advantage	Disadvantage
React	- Component-based architecture. - Virtual DOM to improve performance. - have strong community support.	- Requires an additional state management library (e.g., Redux) - Frequent updates can lead to breaking changes
Angular	- Two-way data binding - Dependency injection - Robust ecosystem and tooling	- Steeper learning curve - Larger bundle sizes - Can be overkill for small projects
Svelte	- No virtual DOM - Highly efficient runtime - Less boilerplate code - Shallow learning curve	- Less mature ecosystem - Limited tooling and community support - Fewer job opportunities

2.5.2 Back-end framework

2.5.2.1 Django

Django is a Python-based framework for building web applications that is both secure and easy to maintain. First released in July 2005, Django is designed to have reusable components and adheres to the ‘Don’t Repeat Yourself’ (DRY) principle to minimize code duplication and enhance maintainability (Rajvanshi & Rajvanshi, 2023).



Figure 2.5.4 Logo of Django

Django operates on the Model-View-Template (MVT) design pattern. The Model handles data, often from a database, using Django's Object-Relational Mapping (ORM) to simplify interactions without needing complex SQL queries. The View manages HTTP requests, processes data with models, and returns the relevant content to the Template, which defines the presentation layer of the application (W3Schools, 2022).

Some of the advantages of Django are that it has strong security measures to prevent such issues as the SQL injection and cross-site scripting. Its “batteries included” philosophy means it provides many built-in functionalities, including user authentication and database management, which are ready to use out of the box (Hamishwillee, 2024).

Django's component-based architecture allows projects to be divided into reusable applications, promoting modularity and scalability. This flexibility can be applied to a variety of web applications including simple content management systems to complex social networks. Also, the use of Python in Django makes it compatible with different operating systems, which is advantageous to developers (Rajvanshi & Rajvanshi, 2023).

Overall, Django offers a full-featured web development framework and focuses on the quality of the code and its security.

2.5.2.2 Laravel

Laravel is PHP framework for web application that was developed in 2011; it is highly valued for its beautiful syntax and the rich set of tools that help to create web applications. It adheres to the Model-View-Controller (MVC) design pattern, which is an effective way of developing web applications (Rajvanshi & Rajvanshi, 2023).



Figure 2.5.5 Logo of Laravel

Laravel enhances developer productivity through its expressive syntax and powerful tools. Key features include a powerful dependency injection system, an rich database abstraction layer, support for queue jobs, support for scheduled jobs, unit testing, and integration testing. These tools facilitate the creation of high-performance, scalable applications (Laravel, 2024).

The framework's scalability is notable, with built-in support for fast caching systems like Redis and platforms such as Laravel Vapor for serverless deployment on AWS. This scalability allows Laravel applications to handle significant traffic and large-scale operations efficiently (Laravel, 2024).

Laravel is equally friendly to beginners and also has a lot of features that are complex for the professional developer. Its progressive nature supports growth, from initial learning to handling complex, enterprise-level projects. Furthermore, Laravel is backed by a large community of developers who are

involved in the continuous development of the framework and provide a rich set of documentation and tutorials (Rajvanshi & Rajvanshi, 2023).

Overall, Laravel is one of the most popular PHP frameworks due to its simplicity, functionality, ability to scale, and active community support.

2.5.2.3 Ruby on Rails

Ruby on Rails (Rails) is a free and open-source web application framework written in Ruby, first released in August 2004. It aims to simplify the development process through its elegant syntax and comprehensive feature set, allowing developers to focus on building applications rather than repeating common tasks. Rails adopts the Model-View-Controller (MVC) architecture, which organizes applications into three components: models handle data and database interactions, views manage the presentation layer, and controllers act as intermediaries, handling user input and coordinating between models and views (Rajvanshi & Rajvanshi, 2023).



Figure 2.5.6 Logo of Ruby on Rails

The framework is built on key principles such as "Don't Repeat Yourself" (DRY), which minimizes code duplication, and "Convention Over Configuration" (CoC), which simplifies development by providing standard conventions. Rails includes built-in tools for tasks like routing, authentication, and asset management, contributing to a streamlined development experience. It also emphasizes security with features that are intended to prevent various threats including XSS and CSRF (Mehra, 2024).

Rails is celebrated for its simplicity and ease of learning, owing to Ruby's readable syntax. It provides robust support for testing through tools like Minitest and RSpec and includes an Asset Pipeline that optimizes assets for better performance. As an open-source framework, Rails is budget-friendly, and its scalability is supported by strong community backing and extensive documentation. This makes Ruby on Rails a versatile and effective choice for web application development, catering to both novice and experienced developers (Rajvanshi & Rajvanshi, 2023).

2.5.2.4 Spring Boot

Java Spring Boot is an open source tool. It is used to make Java-based frameworks easier to use for building web applications and microservices. It is based on Spring framework and provides a clean and modular approach to the development of apps. Spring Boot allows the creation of production-grade, standalone applications that can be run without much configuration, coding of annotations or XML (Microsoft, 2022).



Figure 2.5.7 Logo of SpringBoot

Some of the features of Spring Boot include auto-configuration which automatically configures Spring and third-party libraries based on the project's dependencies and embedded server which allows integration with HTTP servers

like Tomcat and Jetty (Microsoft, 2022). It is an opinionated tool that comes with default dependencies that assist in reducing the build configurations, and the amount of code that needs to be written. Some of the production-grade features that are available in Spring Boot include metrics, health checks, externalized configuration, and so on, which enhance the reliability of the application (Microsoft, 2022).

The CLI tools of Spring Boot also assist in developing and testing applications since it is easy and fast to set up the application. It integrates well with other Spring projects including Spring Data, Spring Cloud, Spring Security and reliable cloud services including Microsoft Azure Spring Cloud (Microsoft, 2022). In general, Spring Boot contributes to reducing the time needed for application development and increasing productivity because of the simplification of creating applications based on Spring (Microsoft, 2022).

Table 2.5 Comparison between Back-end Framework

Back-end Framework	Advantages	Disadvantages
Django	- Integrated components with "batteries included." - Strong security with built-in protections. - Scalable with shared-nothing architecture. - Portable across multiple platforms.	- Steep learning curve. - Monolithic structure may feel rigid. - Can be slower than lightweight frameworks.
Laravel	- Elegant, expressive syntax. - Modular design with robust tools. - Scalable with built-in caching and distributed systems.	- Requires time to master. - Performance can suffer from feature overhead. - PHP's limitations may affect efficiency.

	- Strong community and ecosystem.	
Ruby on Rails	- High productivity with DRY and CoC principles. - Beginner-friendly with readable syntax. - Includes essential features like routing and authentication. - Built-in security protections.	- Convention-heavy, which can limit flexibility. - Performance issues with large apps. - Smaller ecosystem compared to Django and Laravel.
Spring Boot	- Faster Development - Less code - Standalone apps - Easy integration - Production-ready - Developer tools	- Limited customization - Bloated app - Slow startup - Steep learning curve - Complex troubleshooting

2.5.3 Database

2.5.3.1 MySQL

MySQL is a type of open source relational database management system. It is used globally and is owned by Oracle Corporation. MySQL was launched in May 1995 and it is one of the most used databases in the world with over 100 million installations (Sanghvi, 2024). It is well-known for its stability, speed, and simplicity, which is why it is suitable for different web applications and business solutions.



Figure 2.5.8 Logo of MySQL

MySQL is a client-server architecture that employs for data manipulation and data retrieval. It arranges data in tabular form with rows and columns and it follows relational model of data organization. It helps to organize data, search, update and perform various queries in this organization (Talend, 2019). MySQL's attributes include transaction support, multithreaded environment, stored procedures, and a highly developed query optimizer, which make MySQL strong and fast.

The database can be used on a number of computing platforms such as Unix, Windows, and Mac OS. This is because it is an open source application that is developed under GNU General Public License (GPL) hence can be easily customized and interfaced with other systems. For those enterprises requiring the commercial version, Oracle provides a commercially-licensed version (Talend, 2019).

MySQL is one of the most widely used databases in the world with the most popular websites such as Facebook, Twitter, and YouTube using it. It is beginner-friendly as well as suitable for experienced developers, and tools like MySQL Workbench make database management easier (Sanghvi, 2024). The compatibility, relational data management, and backing from the community make MySQL useful in today's big data environment (Talend, 2019).

2.5.3.2 MongoDB

MongoDB is an open source document oriented database which falls under the category of NoSQL databases and is highly scalable and high performance.

Unlike traditional relational databases that store data in tabular format, MongoDB uses a flexible JSON-like format known as BSON (Binary JSON) to store data. This schema-less structure allows for dynamic data storage and retrieval, accommodating diverse and rapidly evolving applications (Sanghvi, 2024).



Figure 2.5.9 Logo of MongoDB

MongoDB excels in handling large volumes of unstructured data and horizontal scalability is achieved through sharding where data is split across multiple servers. This feature makes MongoDB well-suited for applications with significant data needs, such as real-time analytics, user session management, and multimedia content. The database's ability to scale effortlessly and manage extensive workloads contributes to its popularity among major companies like Uber, eBay, Tencent, and The New York Times (Sanghvi, 2024).

Key features of MongoDB include its document-oriented storage, which simplifies data management by storing related data in a single document rather than multiple tables. Indexing improves query efficiency by reducing the need to scan every document. Additionally, MongoDB provides replication and high availability through multiple data copies across different servers, ensuring data redundancy and protection against hardware failures (GeeksforGeeks, 2024).

MongoDB's schema is very flexible and can be updated and changed without affecting the current data, which is very important for applications with dynamic data structures. It has the ability to aggregate data, which translates to performing operations that are similar to SQL's GROUP BYs. In conclusion,

MongoDB is a highly scalable, fast, and easy-to-use database that can effectively address the needs of modern businesses (GeeksforGeeks, 2024).

2.5.3.3 PostgreSQL

PostgreSQL is a robust, open-source object-relational database system renowned for its reliability, data integrity, and extensibility. Originating from the Postgres project initiated in 1986 by Professor Michael Stonebraker, PostgreSQL has evolved significantly, gaining a reputation for handling complex queries and large datasets effectively (Toturiolspoint, 2016).



Figure 2.5.10 Logo of PostgreSQL

As an object-relational database, PostgreSQL extends the traditional relational model with advanced features such as custom data types, operators, and functions. It adheres to ACID (Atomicity, Consistency, Isolation, Durability) principles, ensuring reliability even under heavy workloads. This adherence, combined with performance tuning features like cost-based optimizers, asynchronous replication, and advanced indexing, makes PostgreSQL efficient for diverse applications (Sanghvi, 2024).

PostgreSQL supports complex SQL queries, foreign keys, triggers, and views, and it operates with multiversion concurrency control (MVCC) for high transaction reliability. The database can handle databases up to 32TB and supports various data types, including text, images, and video. Additionally, it offers procedural language support, including PL/pgSQL, PL/Tcl, PL/Perl, and

PL/Python, and can be extended with new data types, functions, and index methods (Toturiolspoint, 2016).

The system is compatible with all major operating systems, including Linux, UNIX, and Windows, and is utilized by major organizations like Reddit, Cisco, and the US Census Bureau. PostgreSQL's combination of standards compliance, extensive features, and proven enterprise scalability continues to drive its adoption across various sectors (Sanghvi, 2024).

2.5.3.4 Oracle

Oracle Database is a commercial enterprise database solution founded in 1977 by Larry Ellison and other developers, which is characterized by high scalability and configurability. It employs the relational model where data is stored in tables and columns with the relationships between the data as objects (SolarWinds, 2024). The current stable release is Oracle 19c which has cloud support and is the first converged, multi-model database. It supports different data types and loads, allows for real-time data processing, and optimizes data ownership, storage, and retrieval (SolarWinds, 2024).

Oracle Database supports multiple operating systems, including Windows and macOS (SolarWinds, 2024). Some of the notable characteristics are auto-tuning of storage, clustering, and memory caching which help in improving performance and dependability. It also provides high availability with Real Application Clusters (RAC) and security features such as Database Vault and Data Safe (SolarWinds, 2024). Also, Oracle offers superior backup and recovery tools, which help to maintain data consistency and reduce the time for data unavailability (SolarWinds, 2024).

In general, Oracle Database is suitable for data warehousing and enterprise grid computing, which enables organizations to store and access various types of data effectively and securely (SolarWinds, 2024).

Table 2.6 Comparison between Database

Database	Advantages	Disadvantages
MySQL	- Mature sharding. - Fast for read-heavy workloads. - Structured schema. - Large community. - Web-friendly.	- Limited complex queries. - Basic transaction support. - Schema changes require migration. - Struggles with large datasets.
MongoDB	- Easy sharding. - High performance. - Flexible schema-less design. - Strong community. - Ideal for evolving data.	- Limited complex transactions. - Potential data issues. - No integrity checks.
PostgreSQL	- Robust scaling. - High performance. - Extensible with custom types. - Strong community. - Multi-OS support.	- Complex setup. - Challenging schema changes. - ACID adds complexity.
Oracle	- Highly flexible. - Diverse data support. - Real-time analysis. - High availability/security. - Strong clustering.	- Difficult setup. - Expensive. - Requires expertise. - Performance overhead. - Costly licensing.

2.6 Conclusion

In conclusion, the existing platforms such as PLAY. FUN, Blinkist, and Yousician are all content-based applications and are built around certain interests. However, they do not offer tailored recommendations based on artificial intelligence as the leading ones do. This results in a unique opportunity

for a platform that uses AI to provide users with personalized recommendations and practical tips on how to improve various interests.

When comparing AI tools, ChatGPT is rather flexible and quite easy to use. Its content generation capability in combination with a simple interface and numerous integration options makes it the best fit for an AI-based platform focused on improving hobbies according to user's preferences.

Regarding the development methodologies, both Agile and DevOps offer great benefits. Both, Agility's concern with responding to new specifications and DevOps' concentration on automation and collaboration are helpful. However, Agile is more suitable for this project due to its iterative nature which is useful in dealing with changing requirements and frequent user feedback.

As for front-end frameworks, Svelte is more performant and has less boilerplate code because of its pre-compilation technique, and hence, it is suitable for performance-intensive frontend development. However, React has more usage and a powerful community, which means that it is more suitable for most projects due to its versatility and solid foundation.

As for back-end frameworks, it is possible to state that both Spring Boot and Laravel have rather powerful solutions. Spring Boot is great for integration and simplicity for Java-based applications while Laravel is easy to use with elegant syntax and comes with a lot of tools for PHP. Based on my choice of Java, Spring Boot would be ideal for this project.

Among MySQL, MongoDB, PostgreSQL, and Oracle, MySQL is well-known for its stability, speed, and large number of users. This is especially useful in web applications that involve a lot of data and are mostly read-intensive. MySQL is still preferred due to its wide compatibility and well-developed ecosystem; however, it is most suitable for those who already have experience with the platform.

CHAPTER 3

METHODOLOGY AND WORK PLANING

3.1 Introduction

In chapter 3 outlines the project's methodology and work planning, providing a structured approach to development. It details how the project will be managed, the development tools and techniques used, and the prototyping and implementation methods. Emphasis is placed on combining agile practices with prototyping methods to ensure iterative development and continuous user feedback. This chapter also presents the work plan, including the tasks and timelines required to achieve the project's goals. A Gantt chart provides a visual representation of the project's progress, illustrating key milestones and deadlines.

3.2 Agile Methodology

The Agile Methodology will be used so as to facilitate the development of iterations and frequent user feedback within the project. Agile emphasizes on flexibility, adaptability and customer satisfaction by providing small incremental improvements in the product. This is important when developing an AI-enhanced platform that is tailored to the user's needs and preferences by responding to feedback and changing needs. The ability to review and refine the development phases at regular intervals will be possible through Agile's iterative cycles known as sprints thus guaranteeing that each phase meets the user and project requirements. When Prototyping is incorporated with Agile, it means that every prototype is developed, tested, and refined in short, easily manageable cycles so that the users are engaged and can help reduce potential significant errors that may be realized later on in the development phase. This not only improves the quality of the product but also the process of development since it becomes more flexible and adaptable. When implemented, Agile will ensure that the project is on the move, and it can overcome any challenge that is likely to arise in the process of developing a user-focused platform that will meet the intended goals of the project.

3.3 Prototyping Methodology

The Prototyping Methodology is selected to work in conjunction with Agile since it facilitates the development of working models that can be easily tested and modified depending on user feedback. This methodology is also consistent with Agile as it involves an iterative approach and the users. Thus, the project can iterate through the ideas very fast and obtain the necessary feedback in the early stages of development. This feedback loop assist in making certain that the end result is both useful and easy to use. The use of Agile and Prototyping makes the project more flexible and able to respond to change as well as make improvements on the project.

3.3.1 Requirement gathering and analysis

As a starting point of the prototyping process, Requirement Gathering and Analysis will be performed with a structure. A Google Form will be used to develop a set of questions that will be used to gather specific requirements from the users and stakeholders. This form will also contain questions that will be aimed at identifying the needs, preferences and challenges that users have when it comes to managing hobbies. The responses will be evaluated based on the number of the issues mentioned and the importance of the requirements in order to make sure that the prototype meets the real-life needs of the target users. In addition to helping to avoid confusion in the process of data collection, this method also gives a clear and logical view of user requirements, which is essential for design and development work.

3.3.2 Quick Design

After the requirement analysis, a Quick Design phase will be introduced, which will provide the overall idea of the prototype. This stage involves creating rough designs of the platform as well as the layout of the user interface based on the collected requirements. The emphasis will be made on the functional design that will represent the main characteristics and the interactions between the user and the system described during the requirement gathering stage. This is a preliminary design that helps in the development process to understand the user requirements and to proceed with the further designing.

3.3.3 Building Prototype

In the Building Prototype phase, the actual model of the first design will be created. Based on the tools and technologies chosen for this project including the front-end using React and the back-end using Laravel, a functional prototype will be created. This will be a mock up of the final product with some of the features and functionalities to enable the user to have a feel of the final product. The aim is to develop a working copy of the application, with which one can interactively experiment and get feedback from users.

3.3.4 Initial User Evaluation

After the creation of the prototype, an Initial User Evaluation will be carried out to get feedback from the actual users. This evaluation entails showing the prototype to a set of users and then getting their feelings, ideas, and feedback. The feedback will be more about the usability, functionality and more importantly the experience of the users. This step is important in order to determine if there are any problems or if there is room for further development before going to the next level.

3.3.5 Testing

The Testing phase is a significant aspect of the Prototyping Methodology since it checks the functionality of the prototype and its compatibility with users' expectations. The first phase of the testing will be done on the prototype to determine the various bugs, usability problems, and functional deficiencies that may be present. This phase will include both the automated and the manual testing techniques which will focus on areas such as performance, security and usability. Information gathered from testing will be used to improve the prototype, solve existing problems and improve the quality of the final product. The cyclical process that is followed here enables constant refinements to be made, thus making the application strong and sound.

3.3.6 Refining the Prototype

From the responses gathered during the first assessment, the Refining the Prototype phase shall entail modification and improvement. This entails correcting usability problems, incorporating new functionalities, and enhancing

the general layout of the website according to the users' feedback. The improved prototype will be tested further to verify that the changes made are sufficient to address the feedback and improve the experience of the users.

3.3.7 Product Implementation and Maintenance

Following the completion of the prototype, the project will go to Product Implementation and Maintenance phase. This phase entails placing the improved prototype into the production environment and making it accessible to the users. Maintenance will also be done continuously to fix problems, incorporate changes as well as ensure that the platform is still useful to the users. The future development of the platform will be based on the results of constant monitoring and feedback from users.

3.4 Development Tools

3.4.1 Front-end

For the front-end development of the application, React will be employed. React is a powerful JavaScript library for building user interfaces, known for its component-based architecture which allows developers to create reusable and modular UI components. By leveraging React, the project can achieve a dynamic and responsive design, offering a seamless user experience. React's efficient rendering and state management capabilities ensure smooth performance, even with complex interfaces. Additionally, its rich ecosystem of libraries and tools will support the development of interactive features and maintainable code.

3.4.2 Back-end

The back-end of the application will be developed using Spring Boot, a popular Java-based framework. Spring Boot simplifies the creation of robust and scalable server-side applications by offering auto-configuration, embedded servers, and a wide range of built-in features. It adheres to a convention-over-configuration approach, which minimizes the need for boilerplate code and accelerates development. Spring Boot's support for dependency injection and

microservices architecture ensures that the application is both modular and easy to maintain, facilitating efficient handling of business logic and data processing.

3.4.3 Database

MySQL will be used for managing the application's data. As a well-established relational database management system, MySQL offers high performance, reliability, and ease of use. Its support for SQL enables efficient data querying and management, while its robust indexing and transaction processing features ensure data integrity and fast access. MySQL's broad compatibility with various platforms and its extensive documentation make it a suitable choice for handling structured data and supporting the application's backend needs.

3.4.4 AI Integration

For AI integration, ChatGPT will be utilized to provide personalized recommendations and interactive features. ChatGPT's advanced natural language processing capabilities allow it to understand and generate human-like responses, making it ideal for enhancing user interactions and delivering tailored suggestions. By integrating ChatGPT, the platform can offer dynamic, AI-driven insights and support, enriching the user experience with personalized guidance and engagement.

build systems (such as Maven or Gradle) which will streamline the process of coding, testing, and deploying your back-end services. Its plugins and tools can help with managing dependencies, setting up servers, and integrating with other tools like databases

3.5.3 Node.js



Figure 3.5.3 Logo of node.js

Node.js will be used to run the development server and manage dependencies for the front-end application. It allows you to use npm (Node Package Manager) to install and manage packages necessary for React development. Node.js also facilitates the use of build tools such as Webpack or Babel, which are essential for optimizing and transpiling your JavaScript code.

3.5.4 IntelliJ IDEA

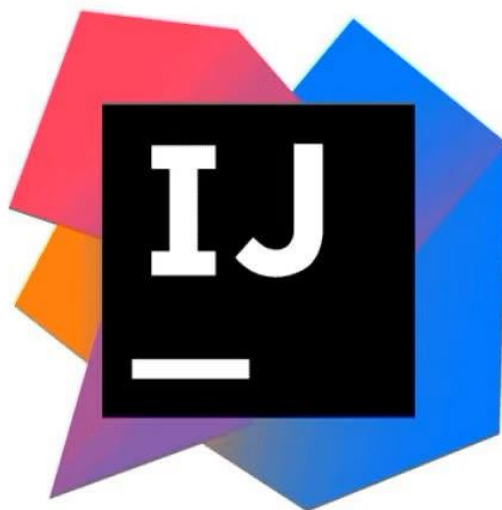


Figure 3.5.4 Logo of IntelliJ IDEA

IntelliJ IDEA is another powerful IDE that supports Java development. It can be used alongside Eclipse for back-end development with Spring Boot. IntelliJ offers advanced code completion, debugging, and testing capabilities. It also integrates well with version control systems, databases, and various development frameworks, providing a seamless development experience.

3.5.5 Postman



Figure 3.5.5 Logo of Postman

Postman will be used for API testing and development. It allows you to create, send, and test HTTP requests to your back-end services. By using Postman, you can ensure that your APIs are working correctly and efficiently. It also provides features for setting up automated tests, which can be used to validate the behaviour of your endpoints and integrations.

3.5.6 Visual Studio Code



Figure 3.5.6 Logo of Visual Studio Code

Visual Studio Code (VS Code) is a lightweight yet powerful source code editor, ideal for front-end development with React. It offers a rich ecosystem of extensions that enhance JavaScript and React development. VS Code provides features like syntax highlighting, IntelliSense, debugging, and Git integration, which streamline the coding process and improve productivity.

3.5.7 Git Bash



Figure 3.5.7 Logo of Git Bash

Git Bash provides a Unix-like command-line interface for Git on Windows. It will be used for executing Git commands, managing version control, and navigating the project repository. Git Bash supports various Git operations, such as cloning repositories, committing changes, and merging branches, which are essential for maintaining code history and collaborating effectively.

3.5.8 WampServer



Figure 3.5.8 Logo of WampServer

WampServer will be utilized for local development and testing of the MySQL database. It provides an environment to run Apache, PHP, and MySQL on Windows. WampServer allows you to create and manage databases, which are crucial for storing user data and application state. It also provides tools for testing SQL queries and ensuring database connectivity.

3.5.9 PhpMyAdmin



Figure 3.5.9 Logo of phpMyAdmin

PhpMyAdmin will be used in conjunction with WampServer to manage the MySQL database through a graphical interface. It simplifies database operations such as creating tables, running queries, and managing user permissions. phpMyAdmin offers an intuitive way to interact with your database, making it easier to perform complex database operations without needing extensive SQL knowledge.

3.6 Work Plan

3.6.1 Work Breakdown Structure(WBS)

0.0 AI-Enhanced Platform for Personalized Hobby Improvement

1.0 Project Planning

1.1 Identify Project Background

1.2 Identify Project Motivation

1.3 Define Problem Statement

1.4 Define Project Objectives

1.5 Define Project Scopes

1.5.1 Identify Target User

1.5.2 Identify System Functionality

1.5.3 Identify Out-of-Scope

1.6 Define Project solution

2.0 Literature Review

2.1 Identify Introduction

2.2 Literature Review on Similar Existing Application

2.2.1 Review PLAY.FUN App

2.2.2 Review Blinkist

2.2.3 Review Yousician

2.3 Review AI-Enhanced Platform

2.3.1 Review ChatGPT

2.3.2 Review Gemini

2.3.3 Review Copilot

2.4 Literature Review on Software Development Methodologies

2.4.1 Review Agile Software Development Methodology

2.4.2 Review Waterfall Software Development Methodology

2.4.3 Review Rapid Software Development Methodology

2.4.4 Review DevOps Software Development Methodology

2.5 Literature Review on Software Development Tools

2.5.1 Review Front-end framework

2.5.1.1 Review React

2.5.1.2 Review Angular

2.5.1.3 Review Svelte

2.5.2 Review Back-end framework

2.5.2.1 Review Django

2.5.2.2 Review Laravel

2.5.2.3 Review Ruby on Rails

2.5.2.4 Review SpringBoot

2.5.3 Review Database

2.5.3.1 Review MySql

2.5.3.2 Review MongoDB

2.5.3.3 Review PostgreSQL

2.5.3.4 Review Oracle

2.6 Requirements Gathering and Analysis

2.6.1 Design Questionnaire

2.6.2 Distribute Questionnaire

2.7 Project scheduling

2.7.1 Create Work Breakdown Structure

2.7.2 Create a Gantt chart

3.0 Methodology and Work planning

3.1 Introduction

3.2 Agile Methodology

3.3 Prototyping Methodology

3.3.1 Requirement gathering and analysis

3.3.2 Quick Design

3.3.3 Building Prototype

3.3.4 Initial User Evaluation

3.3.5 Testing

3.3.6 Refining the Prototype

3.3.7 Product Implementation and Maintenance

3.4 Identify Development Tools

3.4.1 Identify Front-end

3.4.2 Identify Back-end

3.4.3 Identify Database

3.4.4 Identify AI Integration

3.5 Identify Development Utilities

4.0 Quick Design

4.1 Requirements Specification

4.1.1 Identify Functional Requirements

4.1.2 Identify Non-functional Requirements

4.2 Use Case

4.2.1 Design Use Case Diagram

4.2.2 Create Use Case Description

4.3 Design Unified Modeling Language (UML) Diagram

4.4 Define Data Dictionary

4.5 Draw data flow diagram

4.5.1 Draw a Context Diagram

4.5.2 Draw Level 0 Context diagram

4.5.3 Draw Level 1 Context diagram

4.5.3.1 Login and Logout Account

4.5.3.2 SignUp Account

4.5.3.3 Manage profile and account

4.5.3.4 Hobby Recommendation System

4.5.3.5 Search and Discovery Function

4.5.3.6 Content Management

4.5.3.7 Notifications and Alerts

4.5.3.8 Scheduler

4.5.3.9 Admin Dashboard

5.0 Build Prototype

5.1 Create Low Fidelity Prototype

5.1.1 Build Core Functionalities

5.1.2 Design User Interface

5.1.3 Implement Navigation

5.2 Create High Fidelity Prototype

5.2.1 Enhance User Experience

5.2.2 Implement AI Integration

5.2.3 Conduct Usability Testing

5.2.4 Iterate and Refine

6.0 Product Development

6.1 Detailed Design

6.1.1 Define system structure

6.1.2 Design database

6.1.3 Create detailed UI/UX

6.2 Development

6.2.1 Build system

6.2.2 Write code

6.2.3 Review code

6.3 Testing

6.3.1 Test system

6.3.2 Fix issues

6.4 Deployment

6.4.1 Prepare for launch

6.4.2 Set up environment

6.4.3 Launch and test

3.6.2 Gantt Chart



Figure 3.6.1 Project Planning

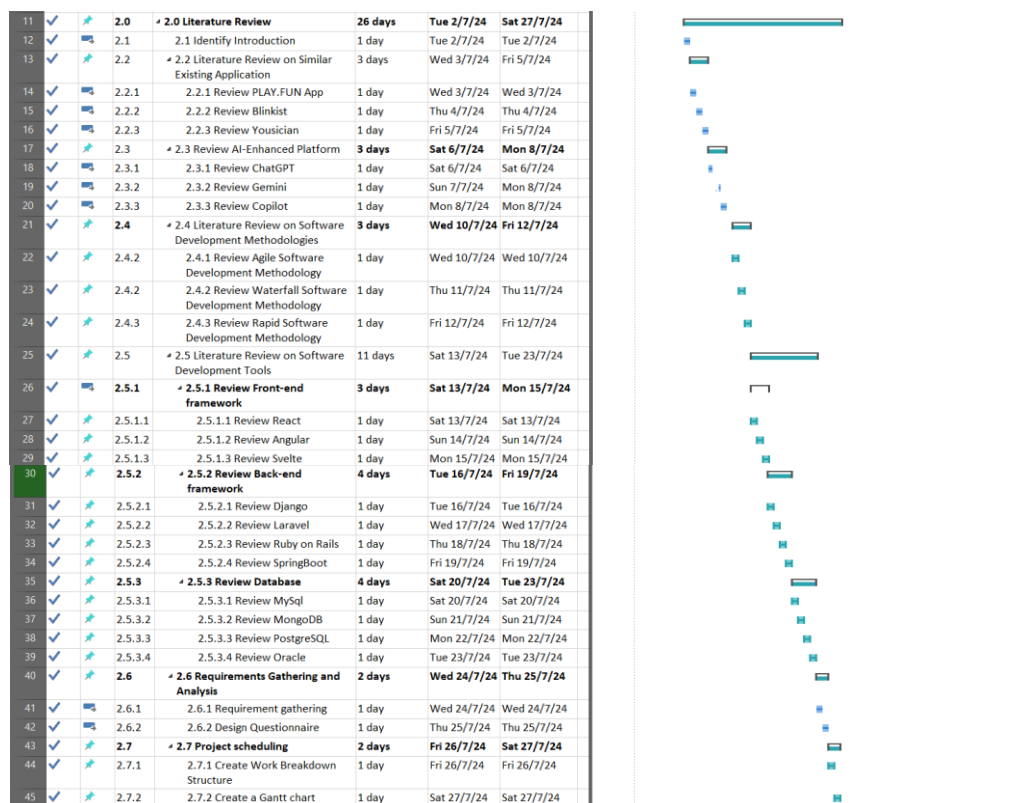


Figure 3.6.2 Literature Review

46	✓	3.0	3.0 METHODOLOGY AND WORK PLAN	16 days	Sun 28/7/24	Mon 12/8/24
47	✓	3.1	3.1 Introduction	1 day	Sun 28/7/24	Sun 28/7/24
48	✓	3.2	3.2 Agile Methodology	3 days	Mon 29/7/24	Wed 31/7/24
49	✓	3.3	3.3 Prototyping Methodology	7 days	Thu 1/8/24	Wed 7/8/24
50	✓	3.3.1	3.3.1 Requirement gathering and analysis	1 day	Thu 1/8/24	Thu 1/8/24
51	✓	3.3.2	3.3.2 Quick Design	1 day	Fri 2/8/24	Sat 3/8/24
52	✓	3.3.3	3.3.3 Building Prototype	1 day	Sat 3/8/24	Sun 4/8/24
53	✓	3.3.4	3.3.5 Initial User Evaluation	1 day	Sun 4/8/24	Sun 4/8/24
54	✓	3.3.5	3.3.5 Testing	1 day	Mon 5/8/24	Mon 5/8/24
55	✓	3.3.6	3.3.6 Refining the Prototype	1 day	Tue 6/8/24	Tue 6/8/24
56	✓	3.3.7	3.3.7 Product Implementation and Maintenance	1 day	Wed 7/8/24	Wed 7/8/24
57	✓	3.4	3.4 Identify Development Tools	4 days	Thu 8/8/24	Sun 11/8/24
58	✓	3.4.1	3.4.1 Identify Front-end	1 day	Thu 8/8/24	Thu 8/8/24
59	✓	3.4.2	3.4.2 Identify Back-end	1 day	Thu 8/8/24	Fri 9/8/24
60	✓	3.4.3	3.4.3 Identify Database	1 day	Fri 9/8/24	Sat 10/8/24
61	✓	3.4.4	3.4.4 Identify AI Integration	1 day	Sun 11/8/24	Sun 11/8/24
62	✓	3.5	3.5 Identify Development Utilities	1 day	Mon 12/8/24	Mon 12/8/24



Figure 3.6.3 Methodology and Work Plan

63	✓	4.0	4.0 Quick Design	19 days	Tue 13/8/24	Sat 31/8/24
64	✓	4.1	4.1 Requirements Specification	2 days	Tue 13/8/24	Wed 14/8/24
65	✓	4.1.1	4.1.1 Identify Functional Requirements	1 day	Tue 13/8/24	Tue 13/8/24
66	✓	4.1.2	4.1.2 Identify Non-functional Requirements	1 day	Wed 14/8/24	Wed 14/8/24
67	✓	4.2	4.2 Use Case	3 days	Thu 15/8/24	Sat 17/8/24
68	✓	4.2.1	4.2.1 Design Use Case Diagram	2 days	Thu 15/8/24	Fri 16/8/24
69	✓	4.2.2	4.2.2 Create Use Case Description	1 day	Sat 17/8/24	Sat 17/8/24
70	✓	4.3	4.3 Database Dictionary	2 days	Sun 18/8/24	Mon 19/8/24
71	✓	4.5	4.5 Draw data flow diagram	3 days	Tue 20/8/24	Thu 22/8/24
72	✓	4.5.1	4.5.1 Draw a Context Diagram	1 day	Tue 20/8/24	Tue 20/8/24
73	✓	4.5.2	4.5.2 Draw Level 0 Context diagram	2 days	Wed 21/8/24	Thu 22/8/24
74	✓	4.5.3	4.5.3 Draw Level 1 Context diagram	9 days	Fri 23/8/24	Sat 31/8/24
75	✓	4.5.3.1	4.5.3.1 Login and Logout Account	1 day	Fri 23/8/24	Fri 23/8/24
76	✓	4.5.3.2	4.5.3.2 SignUp Account	1 day	Sat 24/8/24	Sat 24/8/24
77	✓	4.5.3.3	4.5.3.3 Manage profile and account	1 day	Sun 25/8/24	Sun 25/8/24
78	✓	4.5.3.4	4.5.3.4 Hobby Recommendation System	1 day	Mon 26/8/24	Mon 26/8/24
79	✓	4.5.3.5	4.5.3.5 Search and Discovery Function	1 day	Tue 27/8/24	Tue 27/8/24
80	✓	4.5.3.6	4.5.3.6 Content Management	1 day	Wed 28/8/24	Wed 28/8/24
81	✓	4.5.3.7	4.5.3.7 Notifications and Alerts	1 day	Thu 29/8/24	Thu 29/8/24
82	✓	4.5.3.8	4.5.3.8 Scheduler	1 day	Fri 30/8/24	Fri 30/8/24
83	✓	4.5.3.9	4.5.3.9 Admin Dashboard	1 day	Sat 31/8/24	Sat 31/8/24



Figure 3.6.4 Quick Design

84	✓	5.0	5.0 Build Prototype	43 days	Sun 1/9/24	Sun 13/10/24
85	✓	5.1	5.1 Create Low Fidelity Prototype	3 days	Sun 1/9/24	Tue 3/9/24
86	✓	5.1.1	5.1.1 Build Core Functionalities	1 day	Sun 1/9/24	Sun 1/9/24
87	✓	5.1.2	5.1.2 Design User Interface	1 day	Mon 2/9/24	Mon 2/9/24
88	✓	5.1.3	5.1.3 Implement Navigation	1 day	Tue 3/9/24	Tue 3/9/24
89	✓	5.2	5.2 Create High Fidelity Prototype	40 days	Wed 4/9/24	Sun 13/10/24
90	✓	5.2.1	5.2.1 Enhance User Experience	10 days	Wed 4/9/24	Fri 13/9/24
91	✓	5.2.2	5.2.2 Implement AI Integration	10 days	Sat 14/9/24	Mon 23/9/24
92	✓	5.2.3	5.2.3 Conduct Usability Testing	10 days	Tue 24/9/24	Thu 3/10/24
93	✓	5.2.4	5.2.4 Iterate and Refine	10 days	Fri 4/10/24	Sun 13/10/24

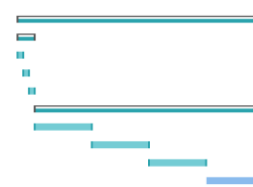


Figure 3.6.5 Build Prototype

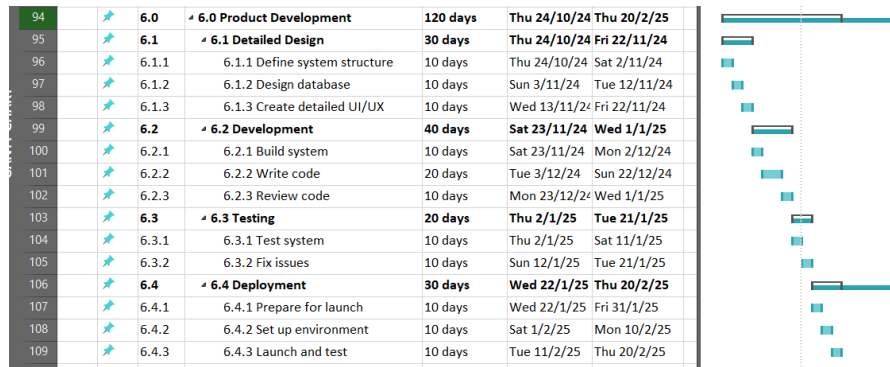


Figure 3.6.6 Product Development

CHAPTER 4

PRELIMINARY RESULTS

4.1 Introduction

This section focuses on assessing the responses to the questions posed in the questionnaire that would help in arriving at the functional and non-functional requirements needed for the creation of the AI-based platform for Personalized Hobby Improvement. According to these requirements, a detailed use case description will be prepared. Also, a Database Dictionary and Data Flow Diagram will be employed for the purpose of illustrating the various capabilities of the system. In order to give a clear picture of how the real system will look and work, a Low Fidelity Prototype will be created.

4.2 Fact Finding

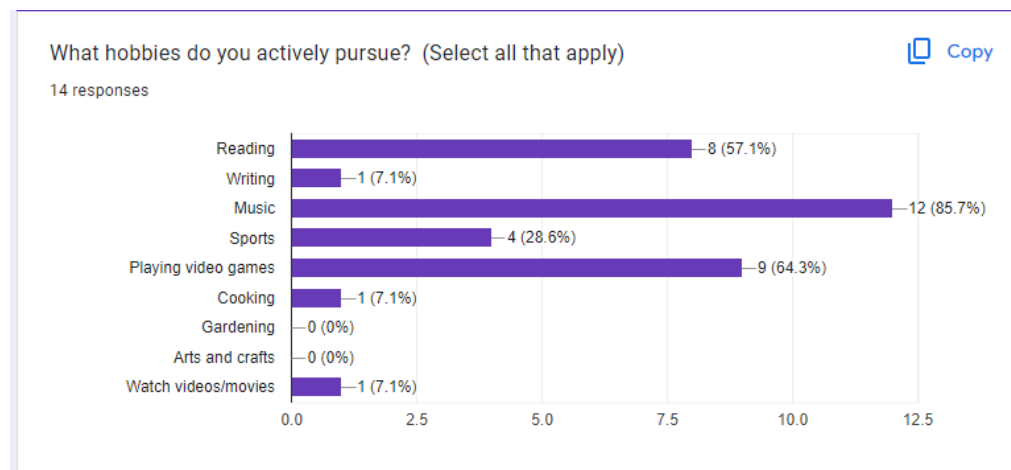


Figure 4.2.1 Hobby pursuits

This question is useful in determining the extent of engagement of people in different hobbies. From the data it is clear that music (85. 7%) and playing video games (64. 3%) are some of the most common hobbies among the respondents while others such as gardening and arts and crafts have no participants at all. It is crucial to comprehend these preferences to adapt the AI-integrated platform to the most popular pursued interests.

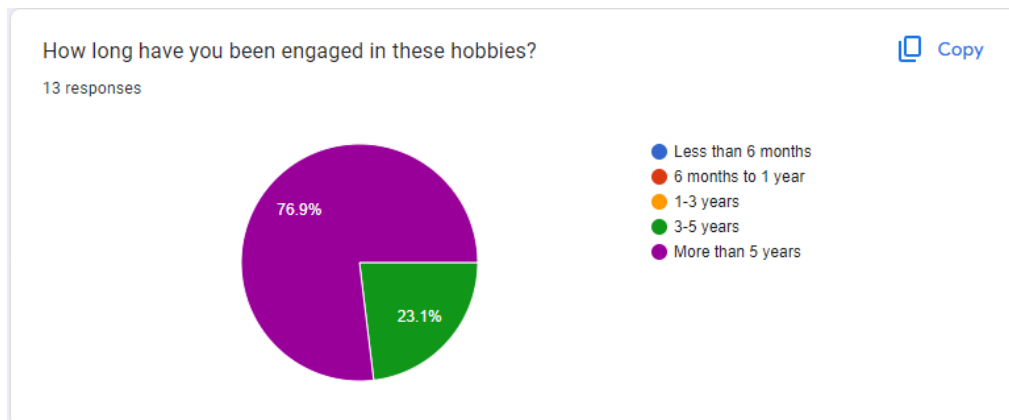


Figure 4.2.2 Time spent doing hobbies

This question informs us how much the participants are conversant with their hobbies. Majority of the respondents (76. 9%) have been practicing their hobbies for 3-5 years and the remaining 23. 1% for more than 5 years. It is also useful to know how long the users spend on the site so as to develop content that is suitable for their experience level.

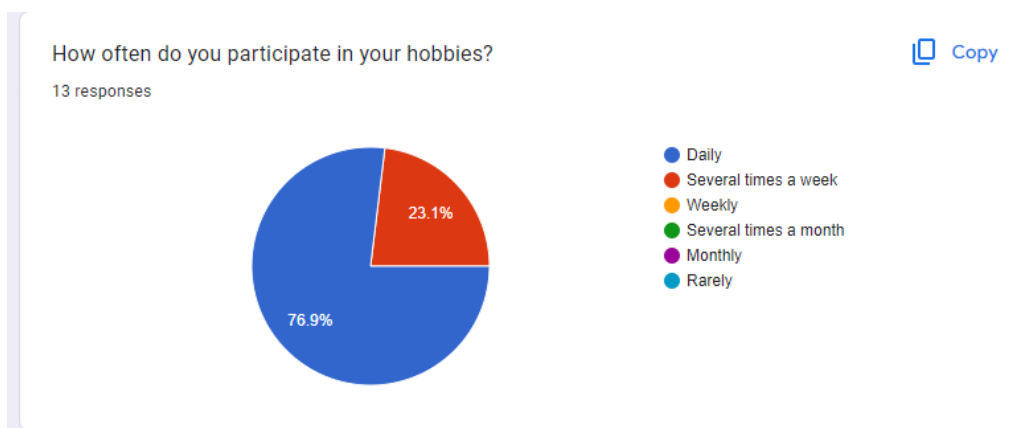


Figure 4.2.3 Number of times engaged in hobbies

This question shows how often people do their leisure activities. Most of them (76. 9%) use it daily, others (23. 1%) use it several times a week. This information is valuable when it comes to developing features that could be incorporated into the user's daily routine.

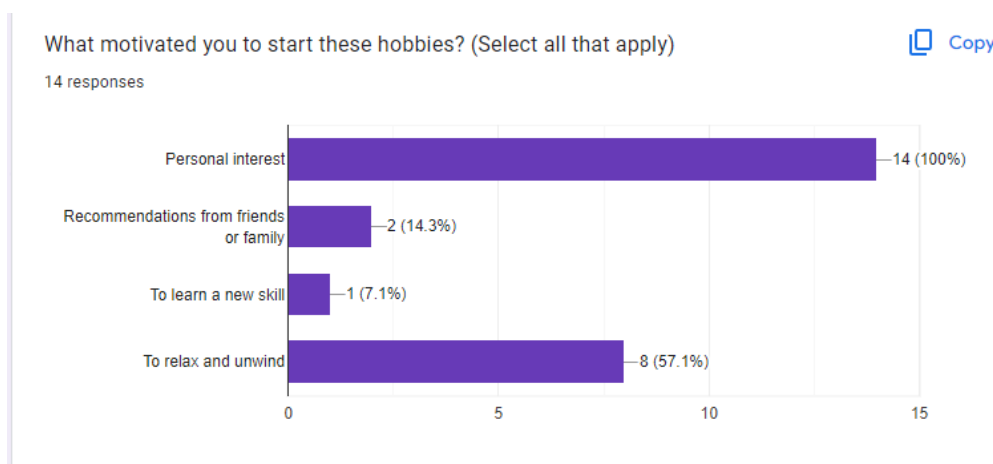


Figure 4.2.4 Reasons for hobbies

This question enables one to know why he or she started a hobby. Personal interest was cited as the primary reason by all the respondents (100%) with others including relaxation (57.1%) and recommendations from other people (14.3%). Understanding these motivations makes it easier in designing experiences that are relevant to the user's preferences and objectives.

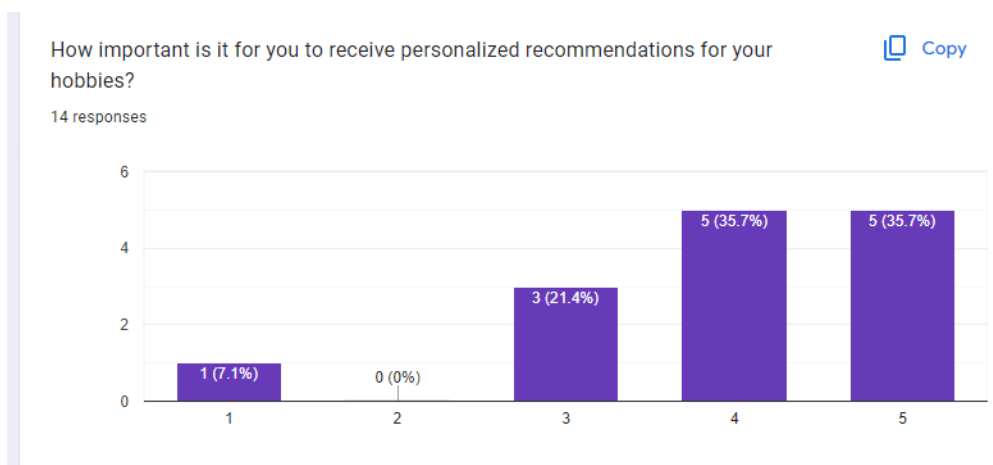


Figure 4.2.5 Importance of personalized recommendations for hobbies

This question aims at determining how much users will be willing to pay for suggestions that are tailored to their needs. This was considered very important by most respondents with 35.7% of the respondents rating it a 5 and 35.7% rating it a 4. This underlines the need for a system that will offer appropriate advice to the users.

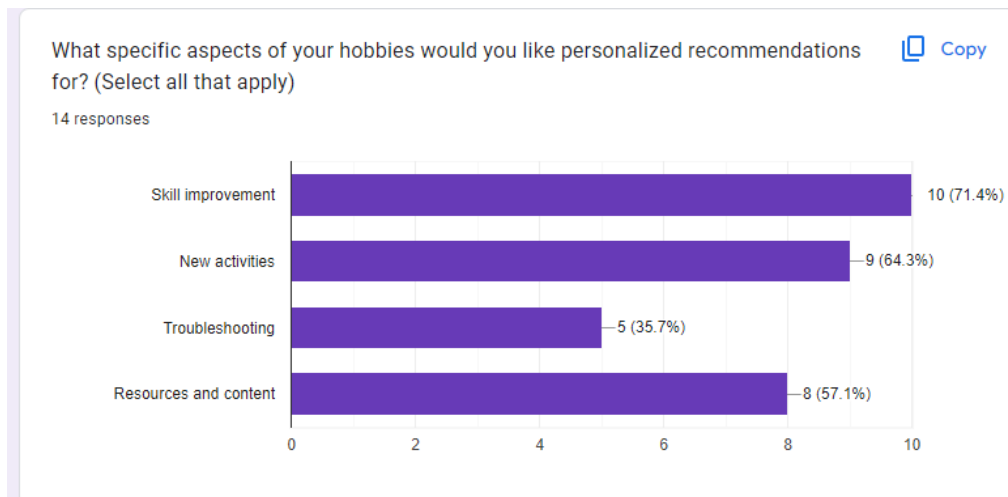


Figure 4.2.6 Aspects targeted by personalized

This question determines where the users need assistance. Among the options given, most of the respondents are interested in suggestions for enhancing the skills (71.4%) and new activities (64.3%) and resources and content (57.1%). Knowledge of these needs is useful in creating features that are based on the improvement of skills and finding new activities.

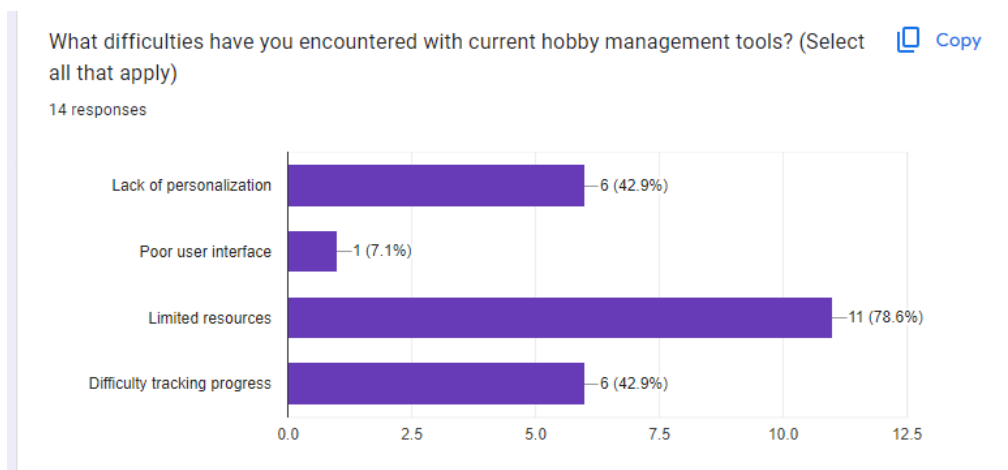


Figure 4.2.7 Difficulties encountered when managing hobbies

This question highlights the problems users face with existing tools. The biggest issue is limited resources (78.6%), followed by lack of personalization (42.9%) and difficulty tracking progress (42.9%). Addressing these pain points will be crucial in the new platform's design.

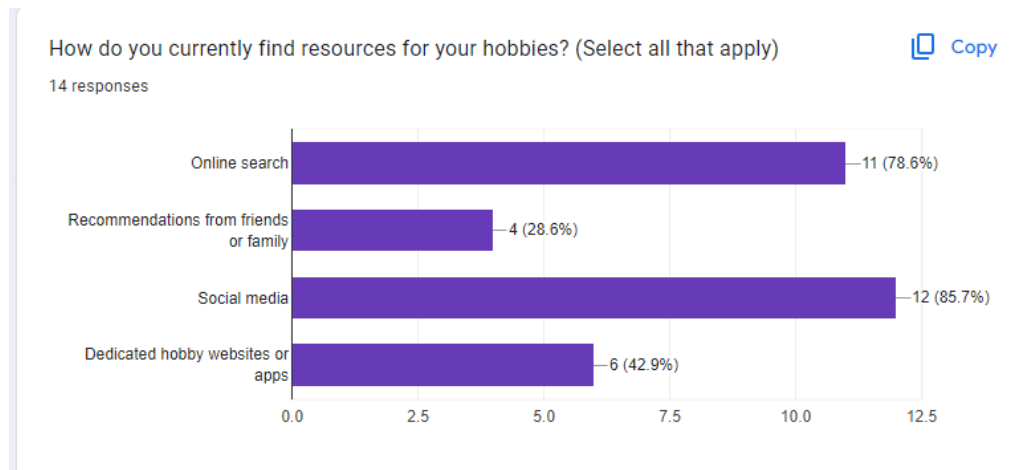


Figure 4.2.8 Resources for hobbies

This question explores how users currently source information for their hobbies. The majority rely on online searches (78.6%) and social media (85.7%). Understanding these habits helps in integrating similar resource-finding features into the platform.

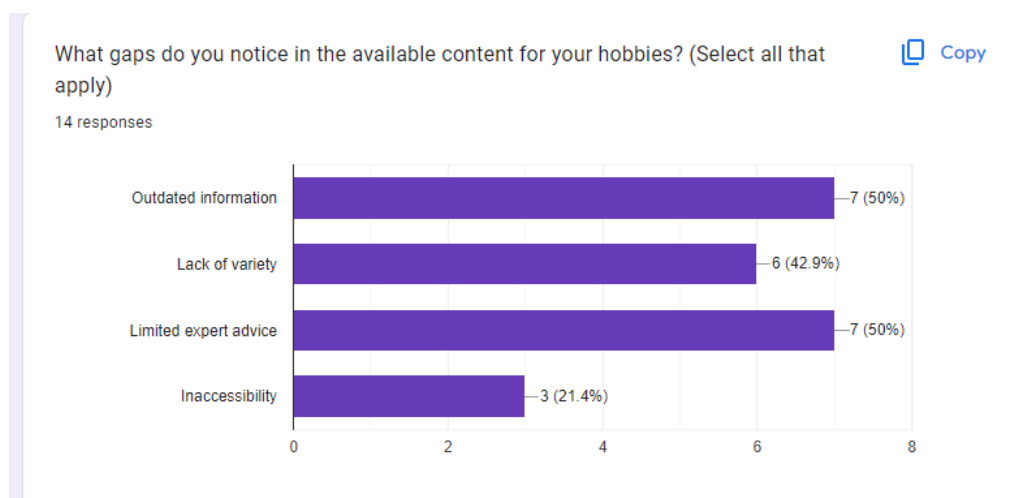


Figure 4.2.9 Gaps in available content for hobbies

This question identifies missing elements in current hobby content. Respondents mentioned outdated information (50%) and limited expert advice (50%) as major gaps. Addressing these gaps can significantly enhance the user experience.

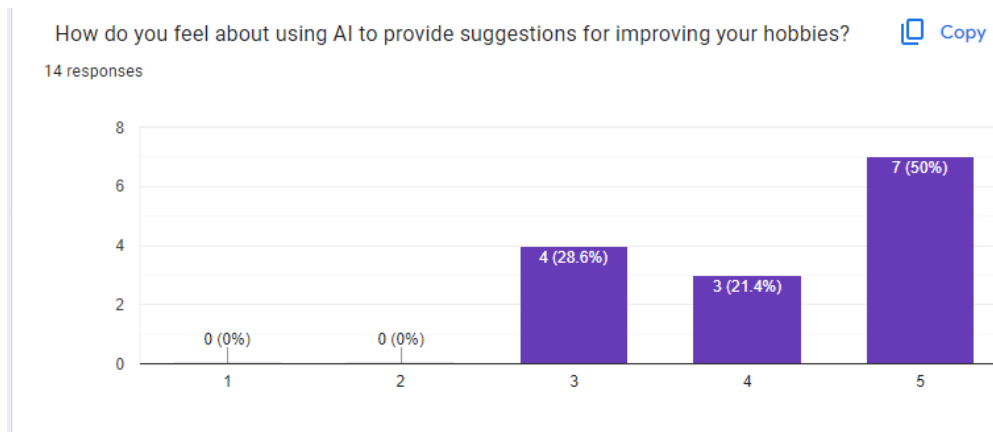


Figure 4.2.10 AI-generated suggestions for improving hobbies

This question aims at determining how much the users are willing to accept advice from an AI. The majority of the respondents have a positive attitude towards the use of AI, 50% of the respondents having given it a 5 while 21.4% a 4. This suggests a high level of support for the integration of AI into the platform.

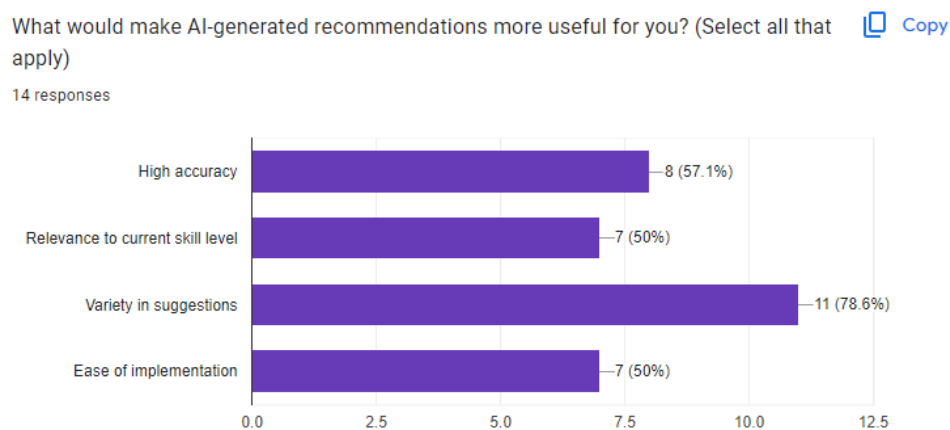


Figure 4.2.11 Willingness to use AI-generated suggestions

This question assists in defining the factors that would make the recommendations provided by AI useful. Users consider high accuracy of the given suggestions (57.1%) and the variety of the offered options (78.6%). This goes to show that it is necessary to make the AI recommendations as specific and varied as possible.

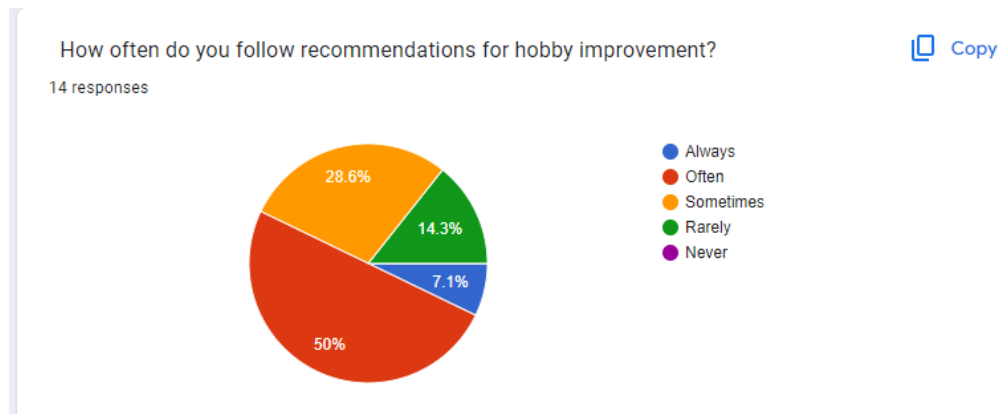


Figure 4.2.12 Number of times suggestions for improving hobbies are followed

This question determines how often the users implement the suggestions. While 50% of the participants said they ‘often’ follow recommendations, a smaller number (7.1%) said they ‘always’ do so, meaning that making recommendations practical and timely can enhance user compliance.

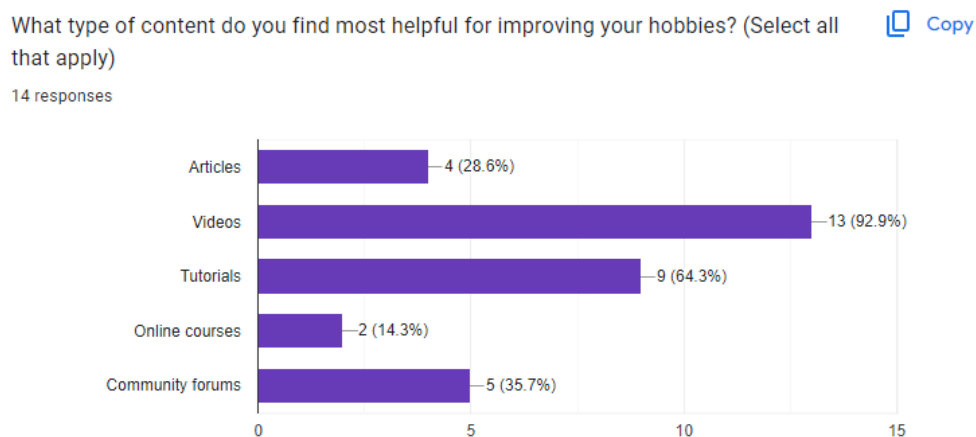


Figure 4.2.13 Most helpful hobby content

This question demonstrates the content type preference. The most used type of content is videos (92.9%), the second most used content type is tutorials (64.3%). The kind of content that the platform should provide can also be informed by this information.

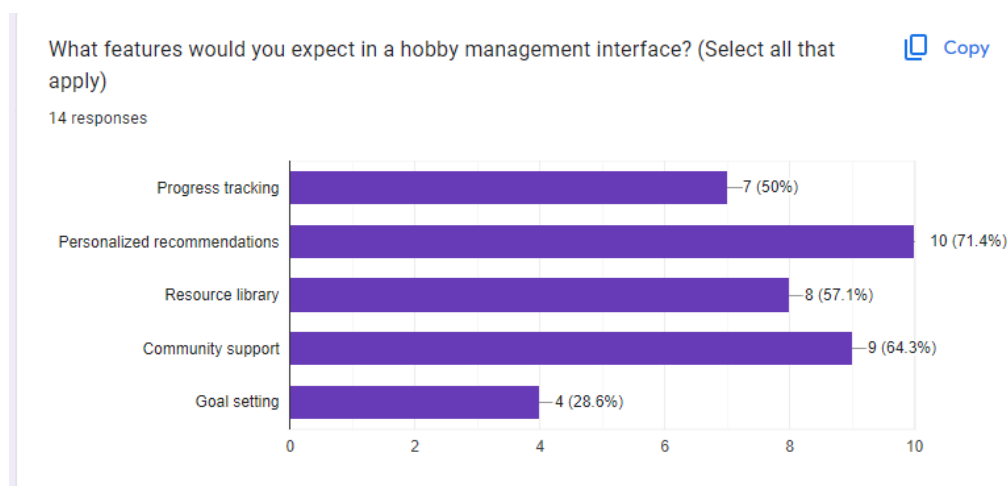


Figure 4.2.14 Features of hobby management interface

This question defines the characteristics that are expected to be present in the platform. Recommendations (71.4%) and community support (64.3%) are the most appreciated, which means that they should be the focus of the platform.

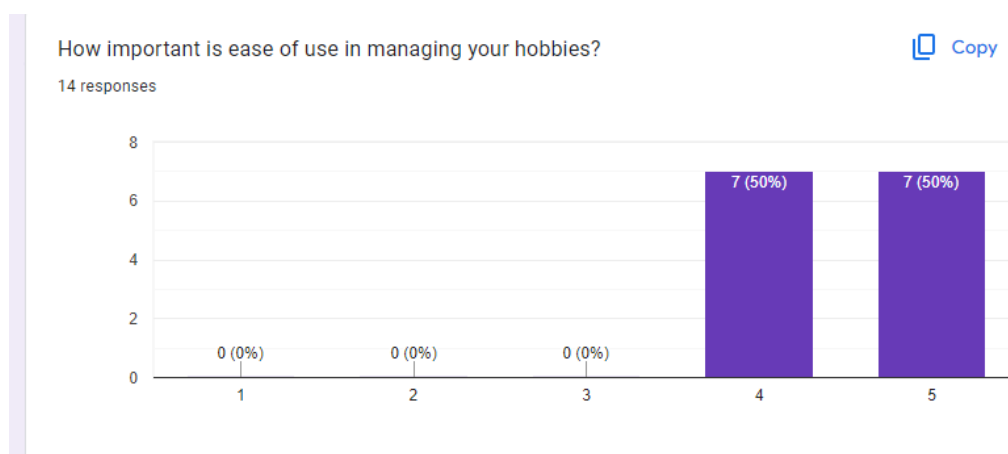


Figure 4.2.15 Importance of ease of use in managing hobbies

This question raises the issue of ease of use design. The importance of ease of use was also considered very high with 50% of the respondents rating it as 5 and the other 50% rating it as 4.

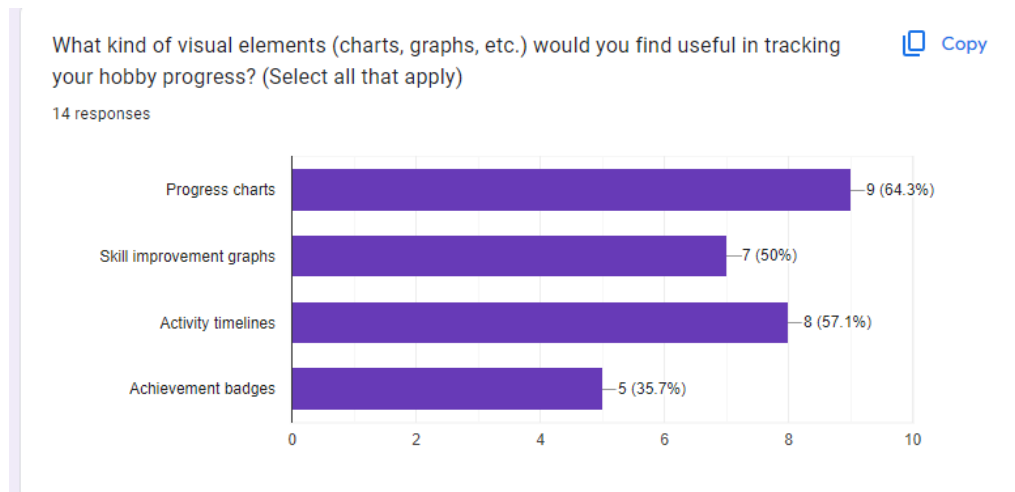


Figure 4.2.16 Visual elements of hobby progress

This question shows the preferred visual aids to use in monitoring progress. The most preferred are progress charts with 64.3% and activity timelines with 57.1% which should be incorporated in the platform design to enable users track their hobby progress.

4.3 Requirement Specification

Based on the information collected from the questionnaire, the following functional and non-functional requirements are specified for the AI-enhanced platform for Personalized Hobby Improvement.

4.3.1 Functional Requirements

Table 4.1 Table of Functional Requirements

ID	Requirements
FR 01	The platform should allow users to manage profiles with details of hobbies, experience, and preferences.
FR 02	The platform should have ways for users to track their progress in different hobbies, such as progress graphs, skill improvement graphs, and activity timelines.
FR 03	The platform should allow users to create goals for their hobbies.
FR 04	The platform should allow users to create milestones for their hobbies.

FR 05	The AI should be able to provide users with personalized recommendations for improving skills, new activities, problems, and resources based on their skill level, interests, and previous activities.
FR 06	Users can also manage the personalized recommendations made to them through a navigation bar.
FR 07	The platform should provide a resource library with articles, videos, tutorials, and online courses related to different hobbies.
FR 08	The platform should have some community support features to enable users to find people with similar interests in hobbies.
FR 09	Users platform be able to share their experiences improving their hobby with other users in the community.
FR 10	The platform should also have a feedback mechanism where users can provide feedback on AI suggestions through ratings.
FR 11	The platform should allow new user to register an account.
FR 12	The platform should allow user to log in and log out an account.
FR 13	The platform should allow admin to manage user accounts , including creating, updating, or deleting user profiles.
FR 14	The platform should allow admin to manage content in the resource library.
FR 15	The platform should allow admin to manage monitor system performance, including user activity, system load, and other metrics.

4.3.2 Non-Functional Requirements

Table 4.2 Table of Non-Functional Requirements

ID	Requirements	Category
NFR 01	The system shall be able to offer suggestions and load resources with the least amount of delay so as not to inconvenience the user.	Performance

NFR 02	The system should be able to respond quickly to user interactions, for instance, in loading content, sending messages or submitting some data.	Performance
NFR 03	The system shall be capable of supporting multiple users simultaneously, and time-sharing shall not have a considerable impact on the system.	Performance
NFR 04	The system shall be highly available, that is, there should be minimal unavailability to allow users access to their hobby management tools and resources at any particular time.	Reliability
NFR 05	In the system, the user data shall be backed up frequently to avoid data loss.	Reliability
NFR 06	The system should be designed to scale for it should be able to handle more users and more resources as the number of users increases.	Scalability
NFR 07	The system shall employ ways and means of securing the users' data such as encryption and secure forms of authentication.	Security
NFR 08	Passwords of users have to be securely hashed before they are stored in the database, and passwords have to be entered in a masked format.	Security
NFR 09	The system shall be user-friendly, with instructions on how to use the system and help links in case the user gets stuck.	Usability
NFR 10	It should be easy to operate, and easy to navigate and the content should be easily understandable by all the users.	Usability
NFR 11	The application should guarantee that the tasks that the users perform are efficient, free of mistakes and do not cause user annoyance.	Usability
NFR 12	The system shall be easily maintainable and the documentation shall be clear so that the system can	Maintainability

4.4.2 Use Case Description

Table 4.3 Use Case Description of Manage User Profile

Name of Use Case : Manage User Profile	ID: UC01	Level of Importance: High
Main Actor: User		
Stakeholders and Benefits: User - Interested in updating and managing their profile, including hobbies, experience, and preferences. Admin - Interested in ensuring users can manage their profiles efficiently. AI - Needs accurate user profile data to provide personalized recommendations.		
Brief Description: The user can update and manage their profile, including hobbies, experience, and preferences. This information is crucial for providing personalized recommendations and tracking progress.		
Trigger: The user chooses to update or manage their profile information from the navigation bar.		
Relationships between Use Case: Association : User Include : Update Profile Details, Set Preferences Extend : Generalization:		
Normal Event Flows: 1. The users login into the platform. 2. The users navigates to the "Manage Profile" section. 3. The users updates their hobbies, experience, and preferences. S-1: Update Profile Details, S-1: Set Preferences 4. The system saves the updated profile details. 5. The users receives confirmation that their profile has been updated successfully.		
Sub-flows: S-1: Update Profile Details 1. The users can update specific details like hobby descriptions, skill levels, and experience. S-1: Set Preferences		

1. The users can adjust their preferences, such as the type of content they prefer and their notification settings.
Flow of Alternative/Exception: 1. If the system encounters an error during a profile update, an error message is displayed.

Table 4.4 Use Case Description of Progress Tracking

Name of Use Case : Manage User Profile	ID: UC01	Level of Importance: High
Main Actor: User		
Stakeholders and Benefits: User - Interested in updating and managing their profile, including hobbies, experience, and preferences. Admin - Interested in ensuring users can manage their profiles efficiently. AI - Needs accurate user profile data to provide personalized recommendations.		
Brief Description: The user can update and manage their profile, including hobbies, experience, and preferences. This information is crucial for providing personalized recommendations and tracking progress.		
Trigger: The user chooses to update or manage their profile information from the navigation bar.		
Relationships between Use Case: Association : User Include : Update Profile Details, Set Preferences Extend : Generalization:		
Normal Event Flows: 6. The users login into the platform. 7. The users navigates to the "Manage Profile" section. 8. The users updates their hobbies, experience, and preferences. S-1: Update Profile Details, S-1: Set Preferences 9. The system saves the updated profile details. 1. The users receives confirmation that their profile has been updated successfully.		

Sub-flows: S-1: Update Profile Details - The users can update specific details like hobby descriptions, skill levels, and experience. S-1: Set Preferences - The users can adjust their preferences, such as the type of content they prefer and their notification settings.
Flow of Alternative/Exception: If the system encounters an error during a profile update, an error message is displayed.

Table 4.5 Use Case Description of Goal Setting

Name of Use Case: Goal Setting	ID: UC03	Level of Importance: Medium
Main Actor: User		
Stakeholders and Benefits: User - Interested in setting and achieving goals for their hobbies. AI - Uses goals to tailor recommendations.		
Brief Description: Users can set specific goals related to their hobbies to track progress and achieve milestones.		
Trigger: The user chooses to set a new goal from the navigation bar.		
Relationships between Use Case: Association : User Include : Extend : Generalization:		
Normal Event Flows: - The users navigates to the "Set Goals" section. - The users defines a goal, including the hobby, target, and timeframe. S-1: Edit Goals - The system saves the goal and adds it to the user's progress tracker.		
Sub-flows: S-1: Edit Goals		

1. The users can modify existing goals.
Flow of Alternative/Exception:

Table 4.6 Use Case Description of Set Milestones

Name of Use Case: Set Milestones	ID: UC04	Level of Importance: Medium
Main Actor: User		
Stakeholders and Benefits: User: Interested in breaking down their goals into smaller, manageable milestones. AI: Uses milestones to track progress and provide targeted recommendations.		
Brief Description: Users can create milestones that serve as checkpoints toward achieving their larger goals within their hobbies.		
Trigger: The users selects the option to set a milestone within the "Goals" section.		
Relationships between Use Case: Association : User Include : Extend : Generalization:		
Normal Event Flows: 1. The users navigates to the "Set Milestones" section. 2. The users selects an existing goal and defines a milestone, including the specific task and target date. S-1: Edit Milestones 3. The platform saves the milestone and links it to the corresponding goal.		
Sub-flows: S-1: Edit Milestones 1. The users can update or delete milestones as needed.		
Flow of Alternative/Exception:		

Table 4.7 Use Case Description of Personalized Recommendations

Name of Use Case: Personalized Recommendations	ID: UC05	Level of Importance: High
Main Actor: AI		
Stakeholders and Benefits: User - Wants relevant and accurate recommendations based on their hobbies, skills, and interests. Admin - Needs to ensure that the AI provides high-quality, accurate recommendations. AI - Must analyze user data and provide personalized recommendations.		
Brief Description: The AI system provides personalized recommendations to users based on their hobbies, skills, and interests, which are tailored to their current skill level, troubleshooting needs, and overall preferences.		
Trigger: The users requests recommendations through the system or interacts with specific features that prompt AI suggestions.		
Relationships between Use Case: Association : AI Include : Tailor to User's Skill Level, Match User's Interests Extend : Troubleshooting Generalization:		
Normal Event Flows: 1. The users login into the platform. 2. The users navigates to the recommendations section. 3. The AI system retrieves the user's profile and activity data. 4. The AI analyzes the user's preferences, skill level, and recent activities. 5. The AI generates personalized recommendations. 6. The users views and interacts with the recommendations. S1: Provide Feedback, S2: Manage Recommendations		
Sub-flows: S-1: Provide Feedback 1. The users provides feedback on the recommendations, indicating their relevance or usefulness.		

S-2: Manage Recommendations
1. The users manages the recommendations by accepting, rejecting, or saving them for later.
Flow of Alternative/Exception:

Table 4.8 Use Case Description of Manage Personalized Recommendations

Name of Use Case: Manage Personalized Recommendations	ID: UC06	Level of Importance: High
Main Actor: User		
Stakeholders and Benefits: User - Wants to receive and manage AI-driven personalized recommendations for improving their hobbies. AI - Continuously refines recommendations based on user feedback and data.		
Brief Description: Users can view and manage personalized recommendations provided by the AI, including suggestions for skill improvement, new activities, and resources.		
Trigger: The users selects the "Manage Personalized Recommendations" option from the navigation bar.		
Relationships between Use Case: Association : User Include : Extend : Provide Feedback Generalization:		
Normal Event Flows: 1. The users navigates to the "Manage Personalized Recommendations" section. 2. The system displays a list of personalized recommendations based on the user's profile, preferences, and activity history. 3. The users reviews the recommendations and decides which ones to follow or ignore. S-1: Provide Feedback 4. The system updates the user's dashboard based on their actions.		
Sub-flows: S-1: Provide Feedback:		

1. The users can rate or provide feedback on the recommendations they received, which is used to improve future suggestions.
Flow of Alternative/Exception:

Table 4.9 Use Case Description of Access Resource Library

Name of Use Case: Access Resource Library	ID: UC07	Level of Importance: High
Main Actor: User		
Stakeholders and Benefits: User - Interested in accessing a wide range of resources to improve their hobbies. Admin - Ensures the resource library is well-maintained and up-to-date.		
Brief Description: Users can browse and search through a library of resources, including articles, videos, tutorials, and online courses related to their hobbies.		
Trigger: The user selects the "Resource Library" option from the navigation bar.		
Relationships between Use Case: Association : User Include : View Resource Extend : Generalization:		
Normal Event Flows: 1. The users navigates to the "Resource Library." 2. The system displays available resources categorized by hobby or type. 3. The users can search for specific resources or browse through categories. S-1: Search advanced function 4. The users selects and accesses a resource.		
Sub-flows: S-1: Search advanced function 1. The users can use the search advanced function to find specific resources quickly.		
Flow of Alternative/Exception: 1. If the search returns no results, the system informs the user and suggests alternative search terms or categories.		

Table 4.10 Use Case Description of Community Support

Name of Use Case: Community Support	ID: UC08	Level of Importance: Medium
Main Actor: User		
Stakeholders and Benefits: User - Interested in connecting with others who share similar hobbies and sharing experiences. Admin - Monitors community interactions to ensure a positive and supportive environment.		
Brief Description: Users can participate in community forums or groups to find support, share experiences, and connect with others who have similar hobbies.		
Trigger: The users selects the "Community Support" option from the navigation bar.		
Relationships between Use Case: Association : User Include : Extend : Share Experiences Generalization:		
Normal Event Flows: 1. The users navigates to the "Community Support" section. 2. The system displays available community groups or forums related to the user's hobbies. 3. The users joins a group or starts a discussion thread. 4. The users shares experiences, asks questions, or offers advice. S-1: Share Experiences		
Sub-flows: S-1: Share Experiences 1. The users can post about their progress or achievements within the community, which others can view and comment on.		
Flow of Alternative/Exception: Inappropriate Content: 1. Admins review and take appropriate action, including removing the content or warning the user.		

Table 4.11 Use Case Description of Share Experiences

Name of Use Case: Share Experiences	ID: UC09	Level of Importance: Low
Main Actor: User		
Stakeholders and Benefits: User - Wants to share their progress, achievements, or challenges with the community. Admin - Monitors shared content to ensure it adheres to community guidelines.		
Brief Description: Users can share their experiences, such as progress in their hobbies or newly learned skills, with the community. Other users can view, comment, and react to these posts.		
Trigger: The user selects the "Share Experiences" option within the "Community Support" section.		
Relationships between Use Case: Association : User Include : Extend : Generalization:		
Normal Event Flows: 1. The users navigates to the "Community Support" section and selects "Share Experiences." 2. The system prompts the user to create a post by describing their experience and optionally attaching images or videos. 3. The users submits the post, and the system adds it to the community feed. S-1: Manage Shared Content 4. Other users can view, comment, and react to the shared experience.		
Sub-flows: S-1: Manage Shared Content 1. The users can edit or delete their post after sharing it.		
Flow of Alternative/Exception: Inappropriate Content: 1. The admin reviews the flagged content and takes appropriate action, such as removing the post or warning the user.		

Table 4.12 Use Case Description of Provide Feedback

Name of Use Case: Provide Feedback	ID: UC10	Level of Importance: Medium
Main Actor: User		
Stakeholders and Benefits: User - Wants to provide feedback on the recommendations they receive to improve future suggestions. AI - Utilizes feedback to refine the recommendation algorithms.		
Brief Description: Users can provide feedback on AI-generated recommendations, which helps the system improve the relevance and accuracy of future suggestions.		
Trigger: The user receives a recommendation and opts to provide feedback on its usefulness or accuracy.		
Relationships between Use Case: Association : User Include : Extend : Generalization:		
Normal Event Flows: 1. The users reviews a personalized recommendation. 2. The users selects the "Provide Feedback" option. 3. The system prompts the user to rate the recommendation and leave comments. 4. The users submits their feedback. S-1: View Past Feedback 5. The system logs the feedback and uses it to adjust future recommendations.		
Sub-flows: S-1: View Past Feedback 1. The users can view feedback they've provided on past recommendations.		
Flow of Alternative/Exception:		

Table 4.13 Use Case Description of Register an Account

Name of Use Case: Register an Account	ID: UC11	Level of Importance: High
Main Actor: User		

Stakeholders and Benefits: New User - Wants to create an account to start using the platform. Admin - Ensures the registration process is smooth and captures all necessary information.
Brief Description: New users can register an account on the platform by providing necessary details such as email, password, and initial preferences.
Trigger: A new user selects the "Register" option on the platform's homepage.
Relationships between Use Case: Association : User Include : Login Account Extend : Generalization:
Normal Event Flows: 1. The new user navigates to the registration page. 2. The system prompts the user to enter their email, create a password, and provide basic profile details. 3. The users submits the registration form. 4. The system creates a new account.
Sub-flows:
Flow of Alternative/Exception:

Table 4.14 Use Case Description of Login Account

Name of Use Case: Login Account	ID: UC12	Level of Importance: High
Main Actor: User		
Stakeholders and Benefits: User - Wants secure access to their account and personalized features. Admin - Ensures that login mechanisms are secure and reliable.		
Brief Description: Users can log into their account to access personalized features, including profile management, progress tracking, and recommendations.		

Trigger: The users selects the "Login" option on the platform's homepage.
Relationships between Use Case: Association : User Include : Extend : Generalization:
Normal Event Flows: 1. The users navigates to the login page. 2. The users enters their username and password. S-1: Password Recovery 3. The system verifies the credentials. 4. If the credentials are correct, the user is granted access to their account dashboard.
Sub-flows: S-1: Password Recovery 1. If the users forgets their password, they can initiate a password recovery process.
Flow of Alternative/Exception: Invalid Credentials: 1. If the users enters incorrect credentials, the system displays an error message.

Table 4.15 Use Case Description of User Management

Name of Use Case: User Management	ID: UC13	Level of Importance: High
Main Actor: Admin		
Stakeholders and Benefits: Admin - Responsible for managing user accounts, including creating, updating, or deleting profiles.		
Brief Description: Admins can manage user accounts by creating, updating, or deleting profiles and ensuring that users have access to the platform's features.		
Trigger: The admin selects the "User Management" option from the navigation bar.		

Relationships between Use Case:		
Association	:	Admin
Include	:	
Extend	:	
Generalization:		
Normal Event Flows:		
1. The admin logs into the platform. 2. The admin navigates to the "User Management" section. 3. The admin views a list of user accounts. 4. The admin selects an account to create, update, or delete.		
S-1: Update User Profile		
5. The system performs the requested action and confirms the update.		
Sub-flows:		
S-1: Update User Profile		
1. The admin can update user information, such as email or access levels.		
Flow of Alternative/Exception:		
User Profile Not Found:		
1. If the admin tries to update or delete a profile that doesn't exist, the system notifies the admin.		

Table 4.16 Use Case Description of Manage Resource Library

Name of Use Case: Manage Resource Library	ID: UC14	Level of Importance: High
Main Actor: Admin		
Stakeholders and Benefits:		
Admin - Responsible for managing the content of the resource library.		
Brief Description:		
Admins can manage the resource library by adding, updating, or removing content to ensure that users have access to relevant and up-to-date resources.		
Trigger:		
The admin selects the "Manage Resource Library" option from the navigation bar.		
Relationships between Use Case:		
Association	:	Admin
Include	:	

Extend : Generalization:
Normal Event Flows: 1. The admin logs into the platform. 2. The admin navigates to the "Manage Resource Library" section. 3. The admin views a list of available resources. 4. The admin selects an option to add, update, or delete a resource. S-1: Update Resource Content 5. The system performs the requested action and confirms the update.
Sub-flows: S-1: Update Resource Content 1. The admin can update the content or metadata of existing resources.
Flow of Alternative/Exception: Resource Not Found: 1. If the admin tries to update or delete a resource that doesn't exist, the system notifies the admin.

Table 4.17 Use Case Description of Monitor System Performance

Name of Use Case: Monitor System Performance	ID: UC15	Level of Importance: Medium
Main Actor: Admin		
Stakeholders and Benefits: Admin - Responsible for ensuring that the platform operates smoothly and efficiently.		
Brief Description: Admins can monitor the system's performance, including server uptime, response times, and user activity levels.		
Trigger: The admin selects the "Monitor System Performance" option from the navigation bar.		
Relationships between Use Case:		
Association : Admin		
Include :		
Extend :		

Generalization:
Normal Event Flows: 1. The admin logs into the platform. 2. The admin navigates to the "Monitor System Performance" section. 3. The system displays performance metrics, including server uptime, response times, and user activity levels. S-1: Generate Performance Reports 4. The admin reviews the metrics and identifies any potential issues.
Sub-flows: S-1: Generate Performance Reports 1. The admin can generate reports based on specific performance metrics for further analysis.
Flow of Alternative/Exception: System Alert: 1. If the system detects performance issues, it automatically alerts the admin.

4.5 Database Dictionary

Table 4.18 Database dictionary for Users table

Field Name	Data Type	Constraints	Description	Relationship
user_id	Integer	Primary	Unique identifier for each user.	One-to-Many with Profiles, HobbyEngagement, Goals, Recommendations
username	String	Unique	User's chosen username for login and identification.	
password_hash	String		Hashed user password for security.	

email	String	Unique	User's email address.	
created_at	Timestamp		Timestamp when the user account was created.	
updated_at	Timestamp		Timestamp of the last update to the user's profile.	

Table 4.19 Database dictionary for Users Profiles table

Field Name	Data Type	Constraints	Description	Relationship
profile_id	Integer	Primary	Unique identifier for each profile.	Many-to-One with Users (user_id)
user_id	Integer	Foreign key references User(user_id)	Link to the Users table.	
hobbies	String		List of hobbies associated with the user.	
experience	String		Description of user's experience in various hobbies.	
preferences	String		User's preferences related to hobbies and activities.	

Table 4.20 Database dictionary for Goals table

Field Name	Data Type	Constraints	Description	Relationship
goal_id	Integer	Primary	Unique identifier for each goal.	Many-to-One with Users (user_id)
user_id	Integer	Foreign key references User(user_id)	Link to the Users table.	One-to-Many with Milestones
goal_description	String		Description of the user's goal for a hobby.	
hobby_name	String		Name of the hobby the goal is related to.	
target_date	Timestamp		The date by which the user aims to achieve the goal.	
status	String		Current status of the goal (e.g., in progress, achieved).	

Table 4.21 Database dictionary for Milestones table

Field Name	Data Type	Constraints	Description	Relationship
milestone_id	Integer	Primary	Unique identifier for each milestone.	Many-to-One with Goals (goal_id)
goal_id	Integer	Foreign key references Goals(goal_id)	Link to the Goals table.	
milestone_desc	String			

			Description of the milestone to be achieved.	
completion_date	Timestamp		Date when the milestone was achieved.	

Table 4.22 Database dictionary for Recommendations table

Field Name	Data Type	Constraints	Description	Relationship
recommendation_id	Integer	Primary	Unique identifier for each recommendation.	Many-to-One with Users (user_id)
user_id	Integer	Foreign key references Goals(goal_id)	Link to the Users table.	
recommendation_text	String		The recommendation provided by the AI.	
hobby_name	String		Name of the hobby related to the recommendation.	
date_created	Timestamp		Date when the recommendation was generated.	

feedback	String		User feedback on the recommendation.	
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Table 4.23 Database dictionary for Recommendations table

Field Name	Data Type	Constraints	Description	Relationship
resource_id	Integer	Primary	Unique identifier for each resource.	
resource_type	Integer		Type of resource (e.g., article, video, tutorial).	
resource_title	String		Title of the resource.	
resource_link	String		URL or location of the resource.	
hobby_name	String		Hobby associated with the resource.	
date_added	Timestamp		Date when the resource was added to the library.	

4.6 Data Flow Diagram

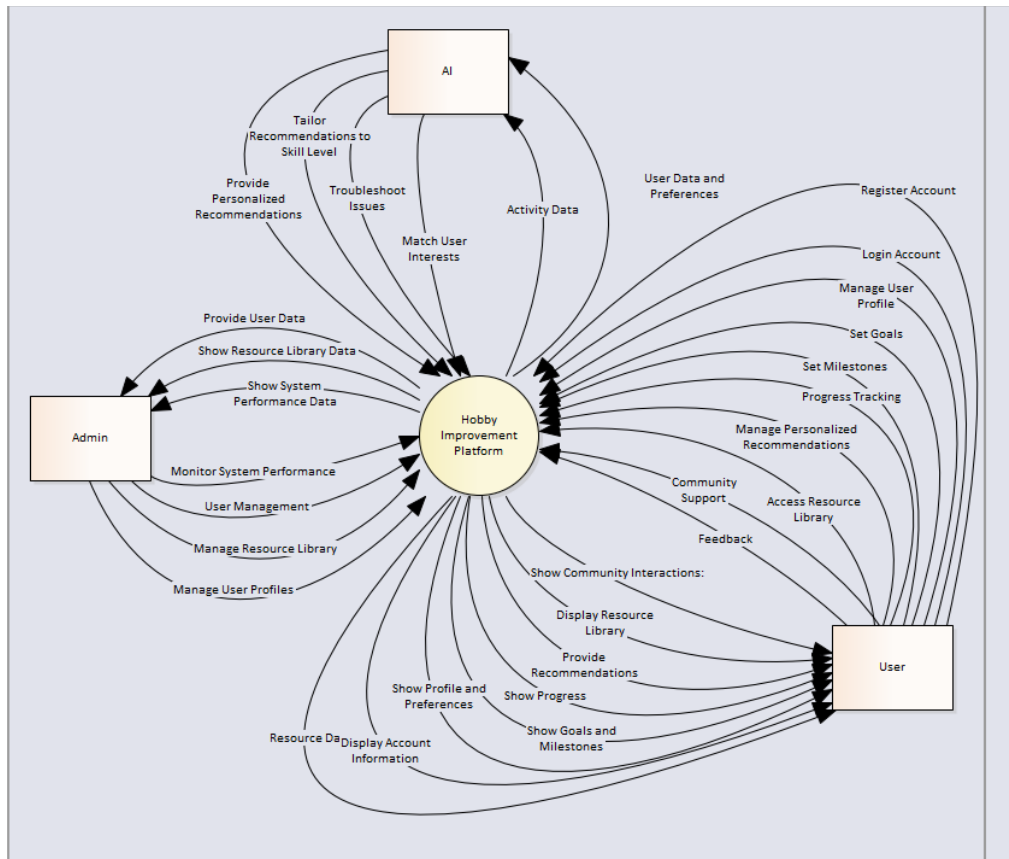


Figure 4.6.1 Figure of Context Diagram

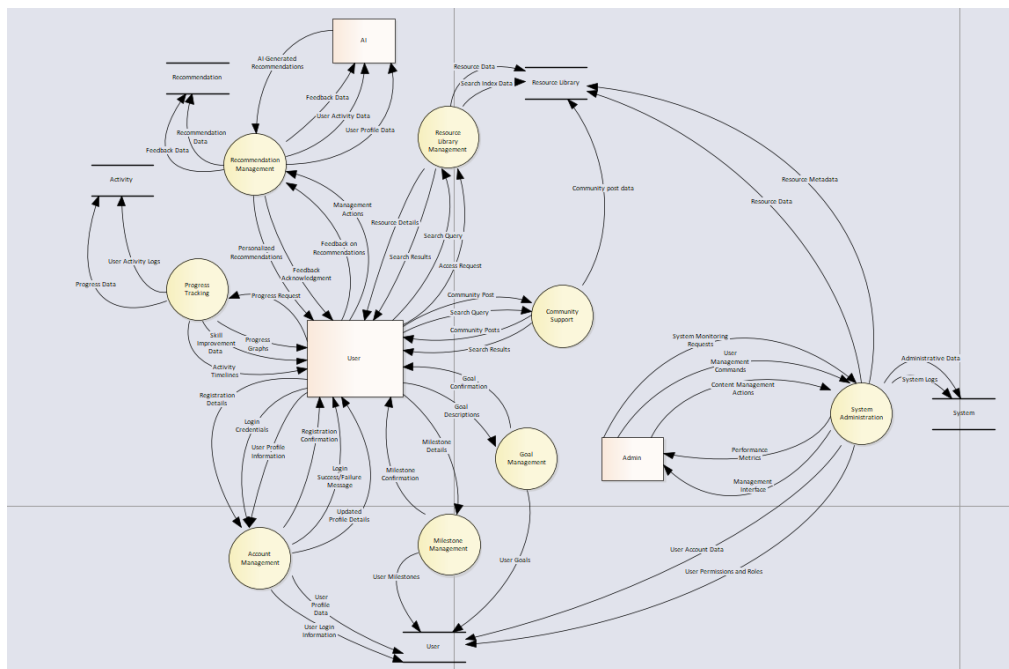


Figure 4.6.2 Figure of Level 0 Data Flow Diagram

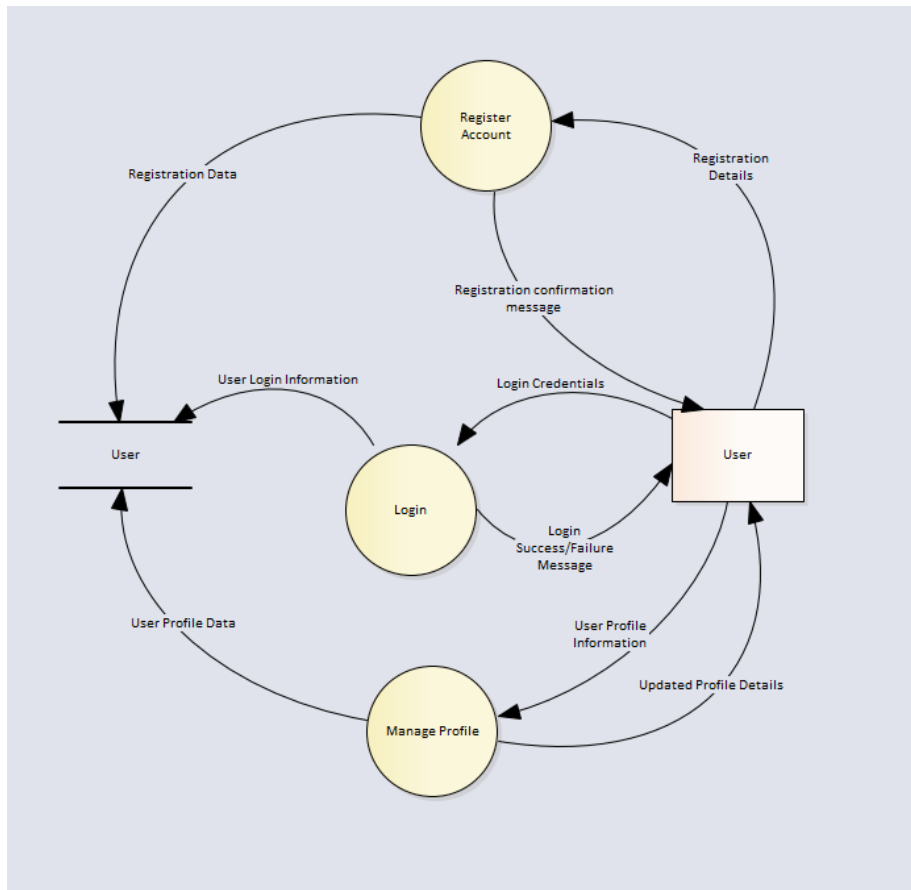


Figure 4.6.3 Account Management

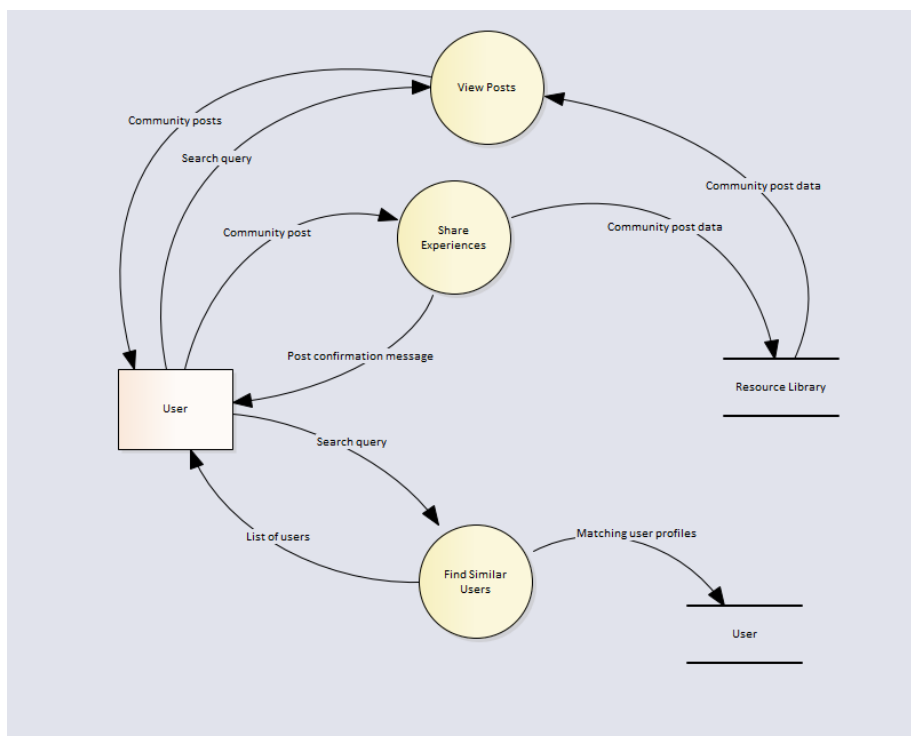


Figure 4.6.4 Community Support

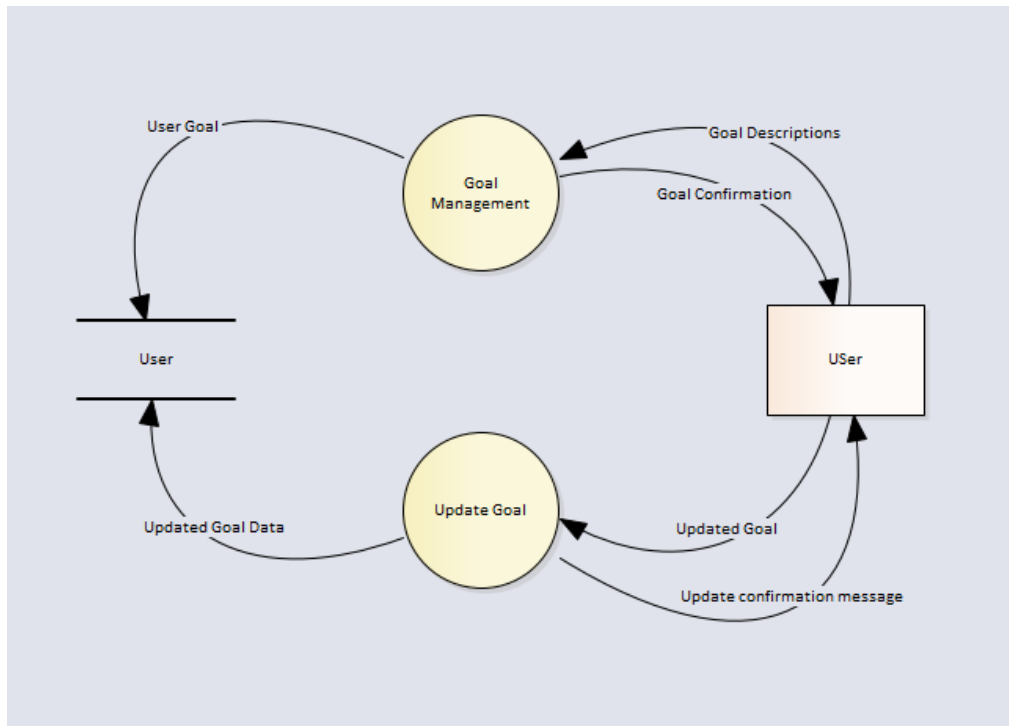


Figure 4.6.5 Goal Management

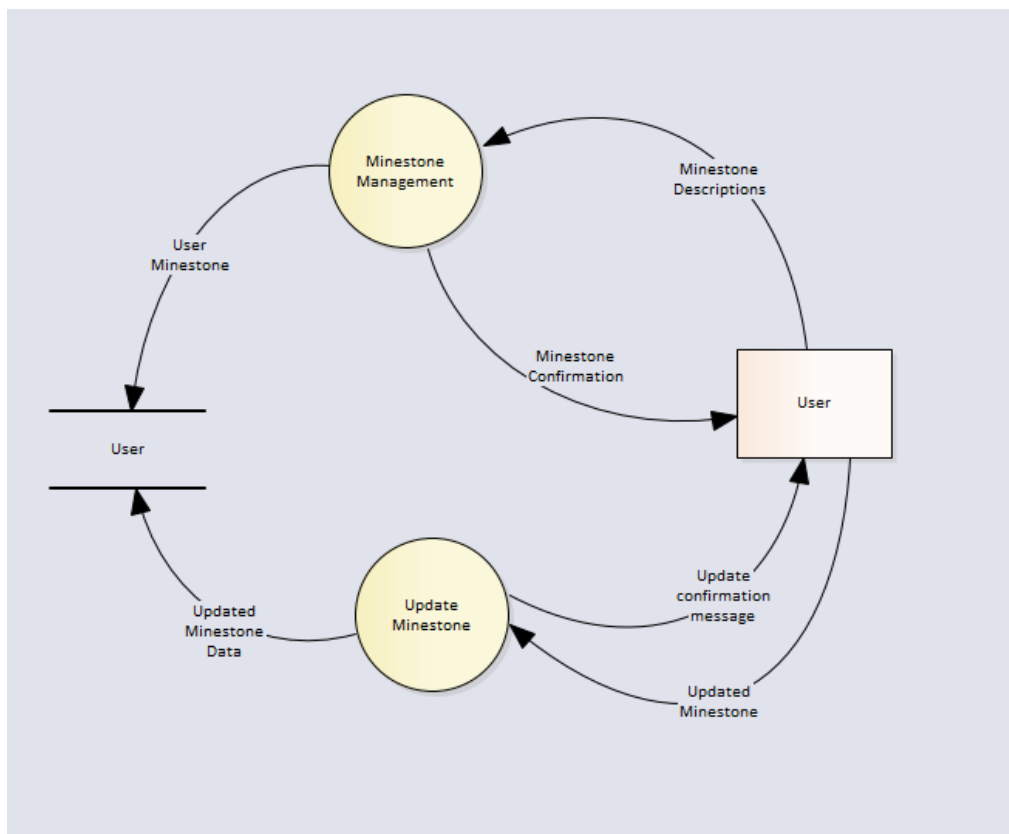


Figure 4.6.6 Milestone Management

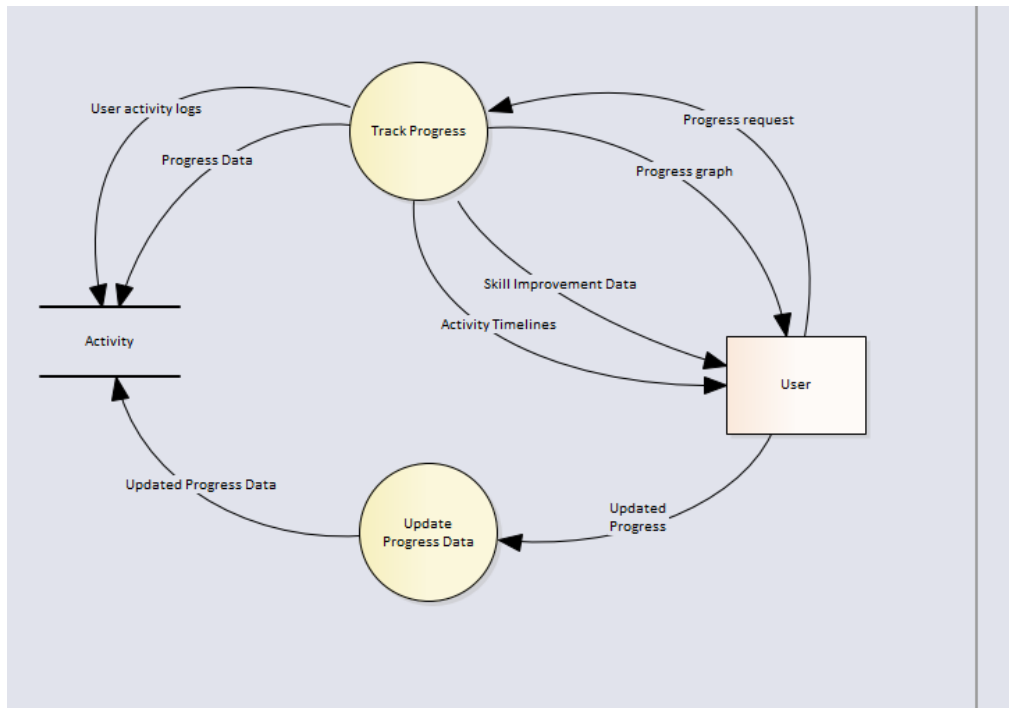


Figure 4.6.7 Progress Tracking

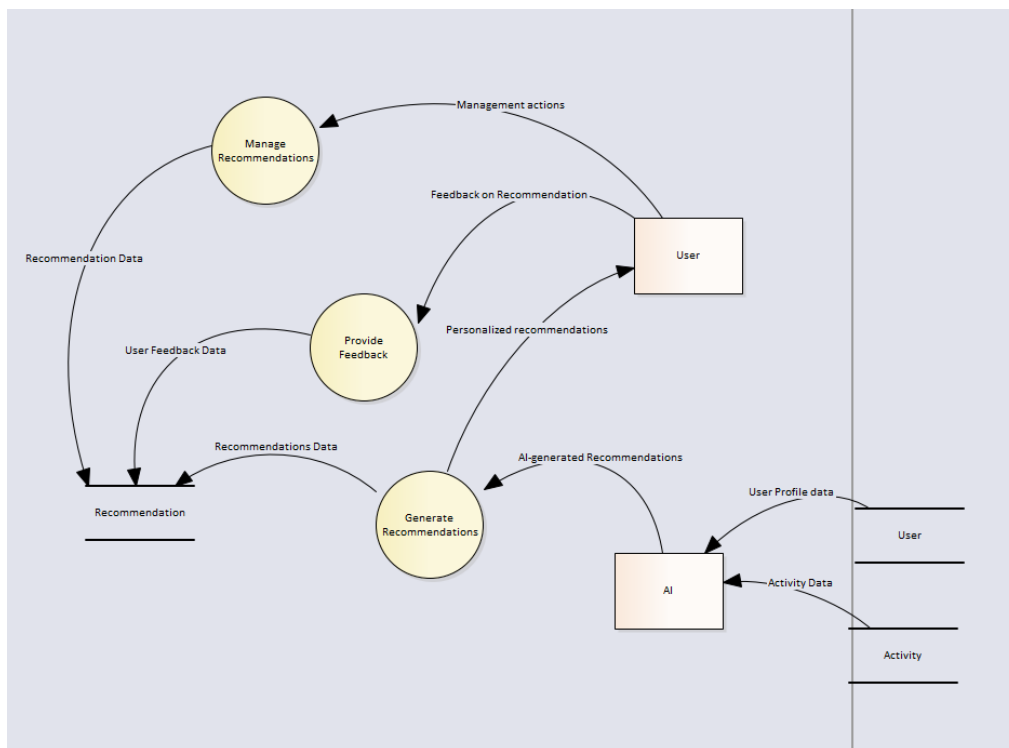


Figure 4.6.8 Recommendation Management

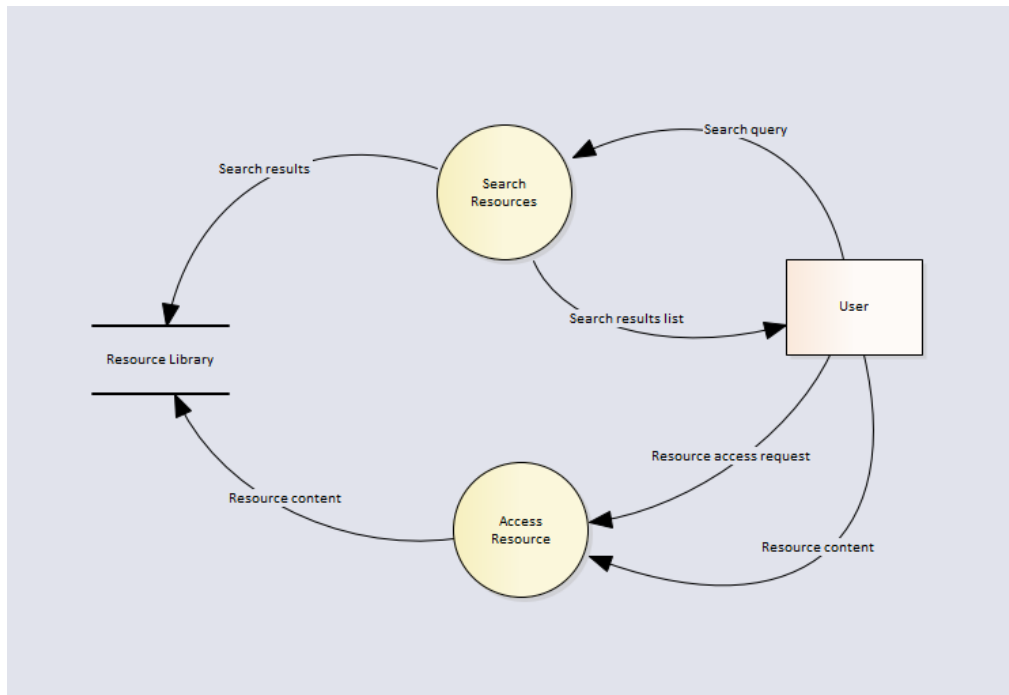


Figure 4.6.9 Resource Library Management

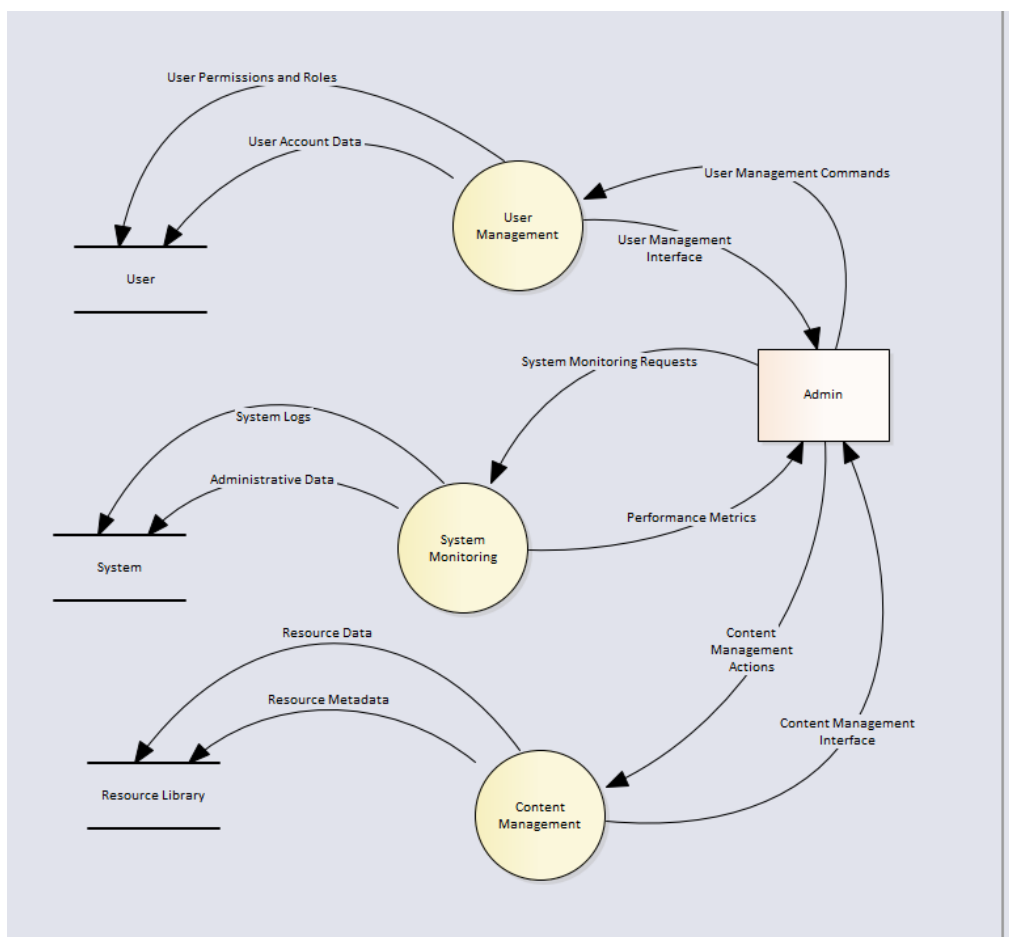
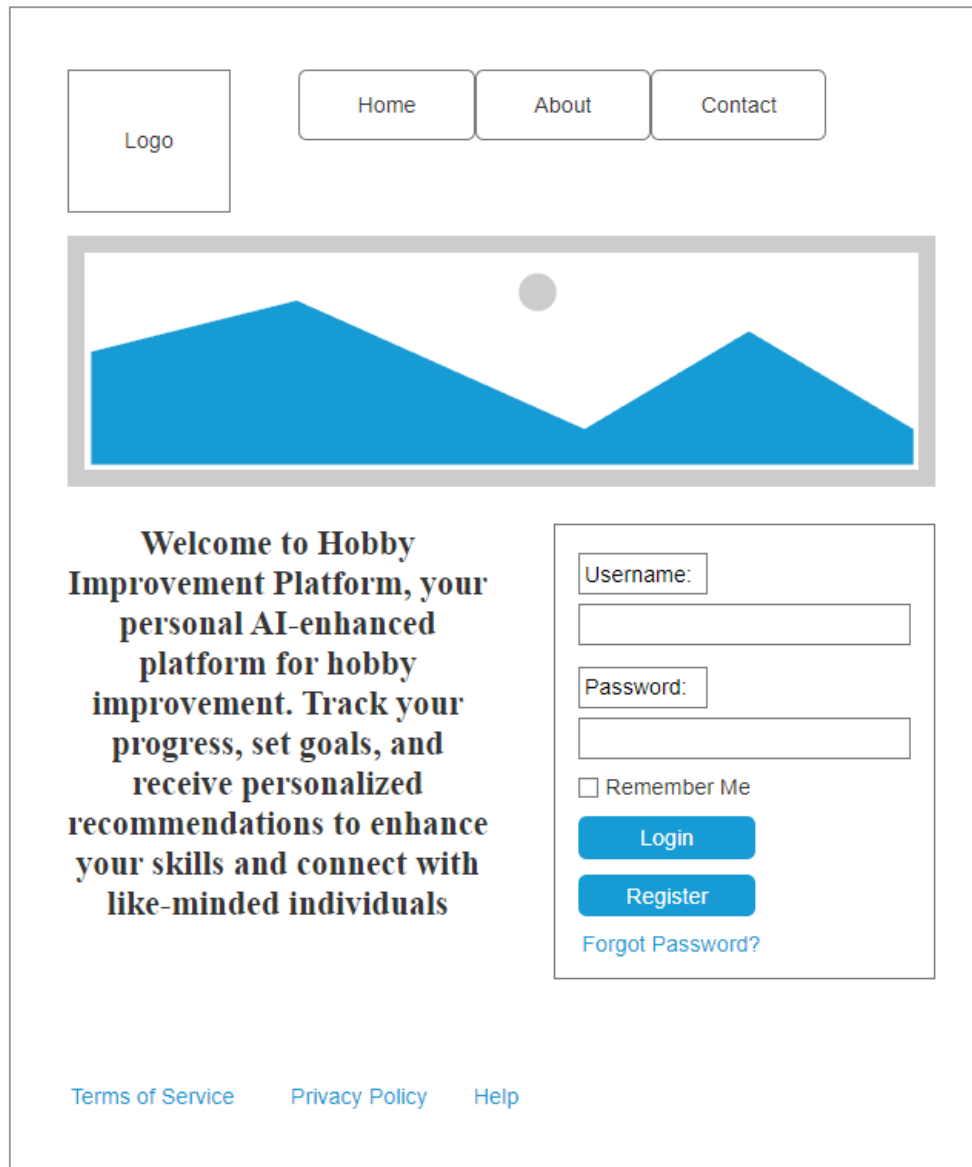


Figure 4.6.10 System Administration

4.7 Low Fidelity Prototype



The image shows a low-fidelity prototype of a web page layout. At the top, there is a navigation bar with a 'Logo' placeholder on the left and three buttons labeled 'Home', 'About', and 'Contact' on the right. Below the navigation bar is a large rectangular area containing a blue abstract graphic with two peaks and a central grey circle. To the left of this graphic, there is a block of bold text: 'Welcome to Hobby Improvement Platform, your personal AI-enhanced platform for hobby improvement. Track your progress, set goals, and receive personalized recommendations to enhance your skills and connect with like-minded individuals'. To the right of the graphic is a login/register form. The form contains two input fields labeled 'Username:' and 'Password:', a checkbox labeled 'Remember Me', two blue buttons labeled 'Login' and 'Register', and a blue link labeled 'Forgot Password?'. At the bottom of the page, there are three links: 'Terms of Service', 'Privacy Policy', and 'Help'.

Logo

Home About Contact

Welcome to Hobby Improvement Platform, your personal AI-enhanced platform for hobby improvement. Track your progress, set goals, and receive personalized recommendations to enhance your skills and connect with like-minded individuals

Username:

Password:

☐ Remember Me

Login

Register

[Forgot Password?](#)

[Terms of Service](#) [Privacy Policy](#) [Help](#)

Figure 4.7.1 Main Home Page

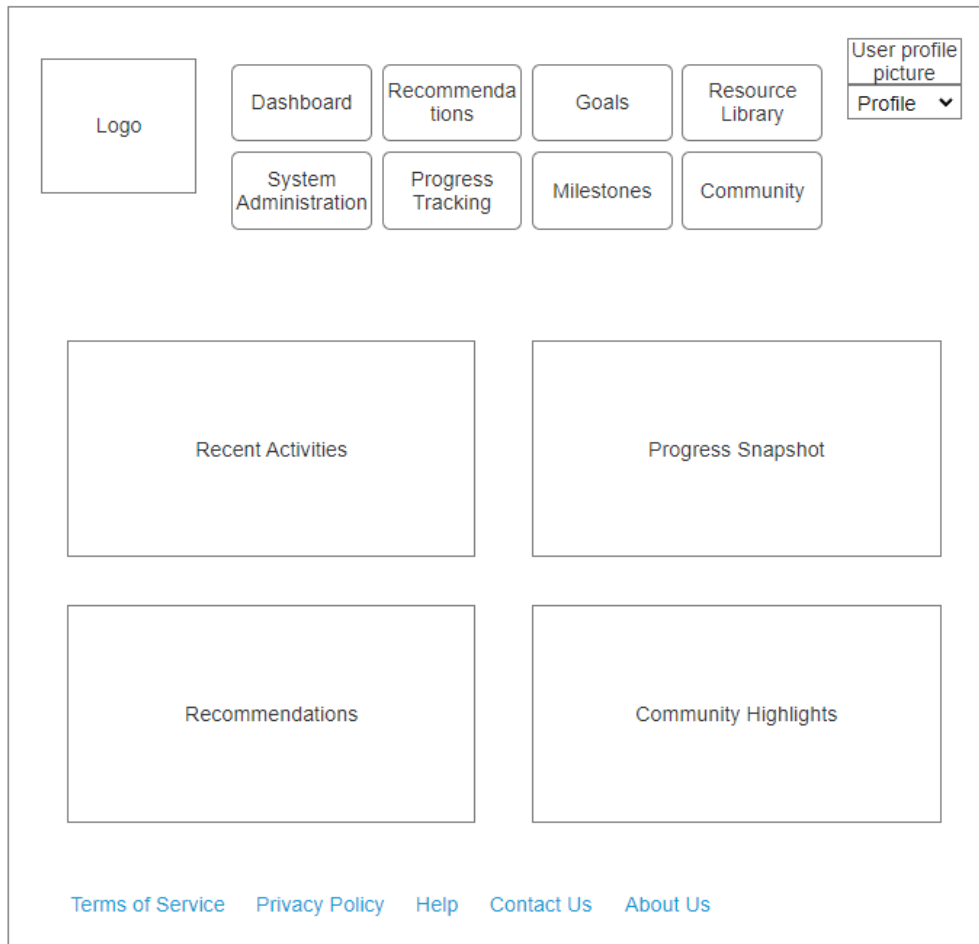


Figure 4.7.2 Post-Login Dashboard Page

Profile Picture

Username:

XXX XXXX XXXXX

Email:

xxxxxxxxx@gmail.com

Phone Number:

+60 ▾

XXX XXXX XXXXX

Password:

XXX XXXX XXXXX

Hobbies:

Reading ▾

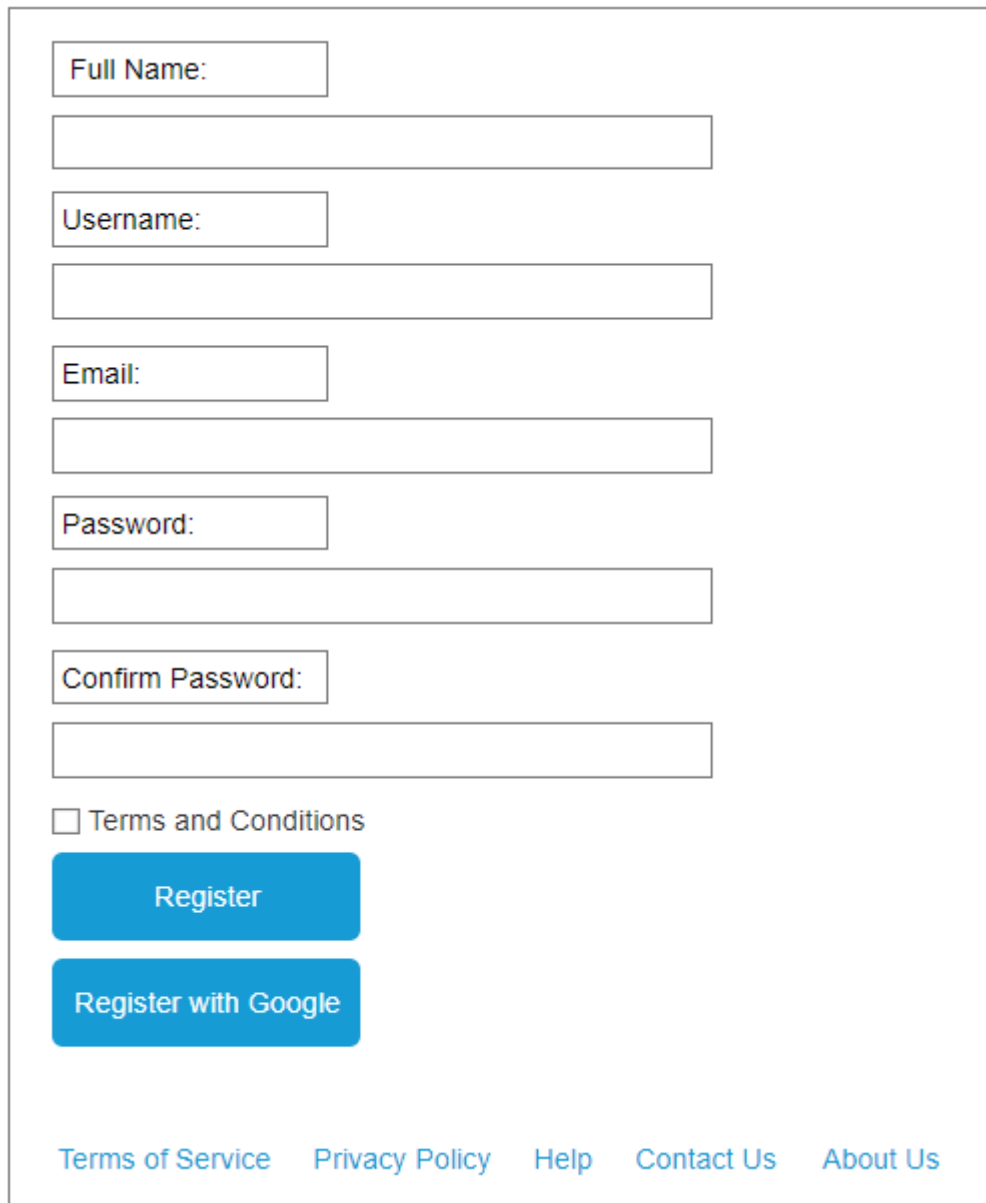
Experience Level:

Beginner ▾

Save and Change

[Terms of Service](#) [Privacy Policy](#) [Help](#) [Contact Us](#) [About Us](#) [Logout](#)

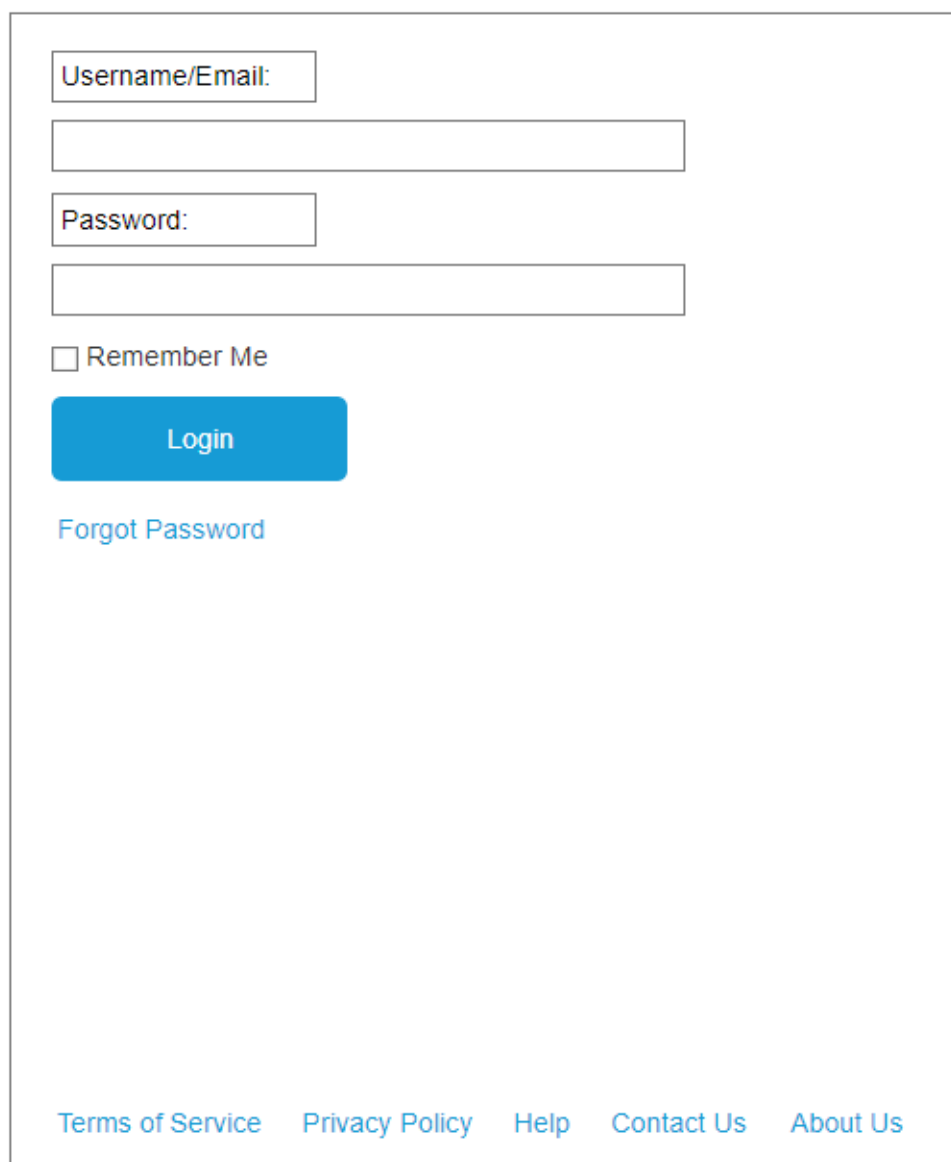
Figure 4.7.3 Profile Management Page



The image shows a registration form with the following elements:

- Full Name:** A label followed by a text input field.
- Username:** A label followed by a text input field.
- Email:** A label followed by a text input field.
- Password:** A label followed by a text input field.
- Confirm Password:** A label followed by a text input field.
- ☐ **Terms and Conditions**
- Register**: A blue button.
- Register with Google**: A blue button.
- [Terms of Service](#) [Privacy Policy](#) [Help](#) [Contact Us](#) [About Us](#): A row of five links at the bottom.

Figure 4.7.4 Register Page



A login page form with a light gray background. It contains two input fields for 'Username/Email' and 'Password', each with a label above it. Below the password field is a checkbox labeled 'Remember Me'. A blue 'Login' button is positioned below the checkbox. A link 'Forgot Password' is located below the button. At the bottom, there is a horizontal row of five links: 'Terms of Service', 'Privacy Policy', 'Help', 'Contact Us', and 'About Us'.

Username/Email:

Password:

☐ Remember Me

Login

[Forgot Password](#)

[Terms of Service](#) [Privacy Policy](#) [Help](#) [Contact Us](#) [About Us](#)

Figure 4.7.5 Login Page

Goal Setting Section

Goal Title:

XXX XXXX XXXXX

Description:

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aenean euismod bibendum laoreet. Proin gravida dolor sit amet lacus accumsan et viverra justo commodo. Proin sodales pulvinar sic tempor. Sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Nam fermentum, nulla luctus pharetra vulputate, felis tellus mollis orci, sed rhoncus pronin sapien nunc accuan eget.

Target Date:

XX/XX/XXXX

Priority Level:

Low ▼

Add new Goal

Save and Change

[Terms of Service](#)
[Privacy Policy](#)
[Help](#)
[Contact Us](#)
[About Us](#)
[Logout](#)

Figure 4.7.6 Goal Page

Milestone Creation

Milestone Title:

XXX XXXX XXXXX

Milestone Deadline:

XX/XX/XXXX

Task list:

xxxx ▼

Save and Change

[Terms of Service](#)
[Privacy Policy](#)
[Help](#)
[Contact Us](#)
[About Us](#)
[Logout](#)

Figure 4.7.7 Milestone Page

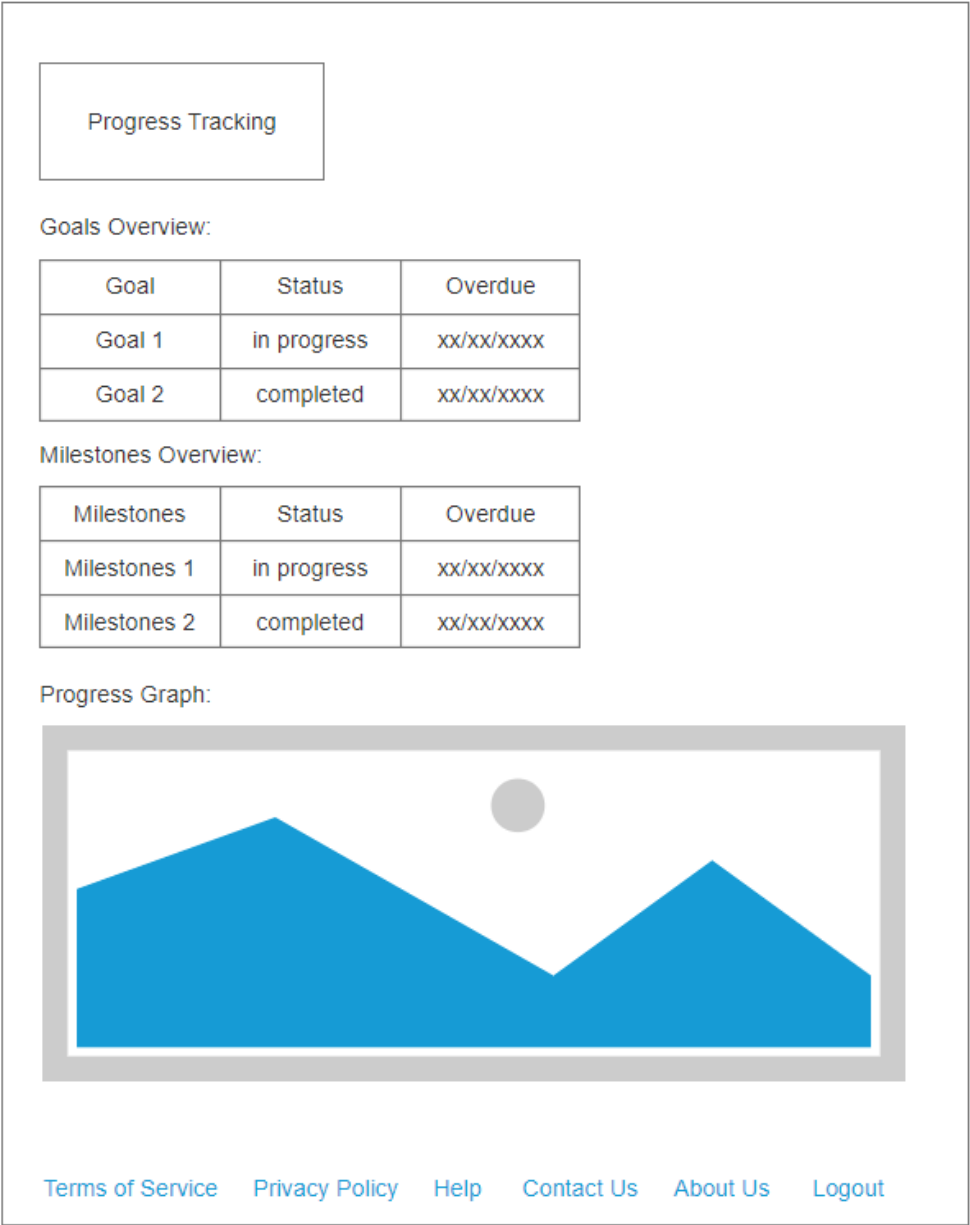


Figure 4.7.8 Progress Tracking Page

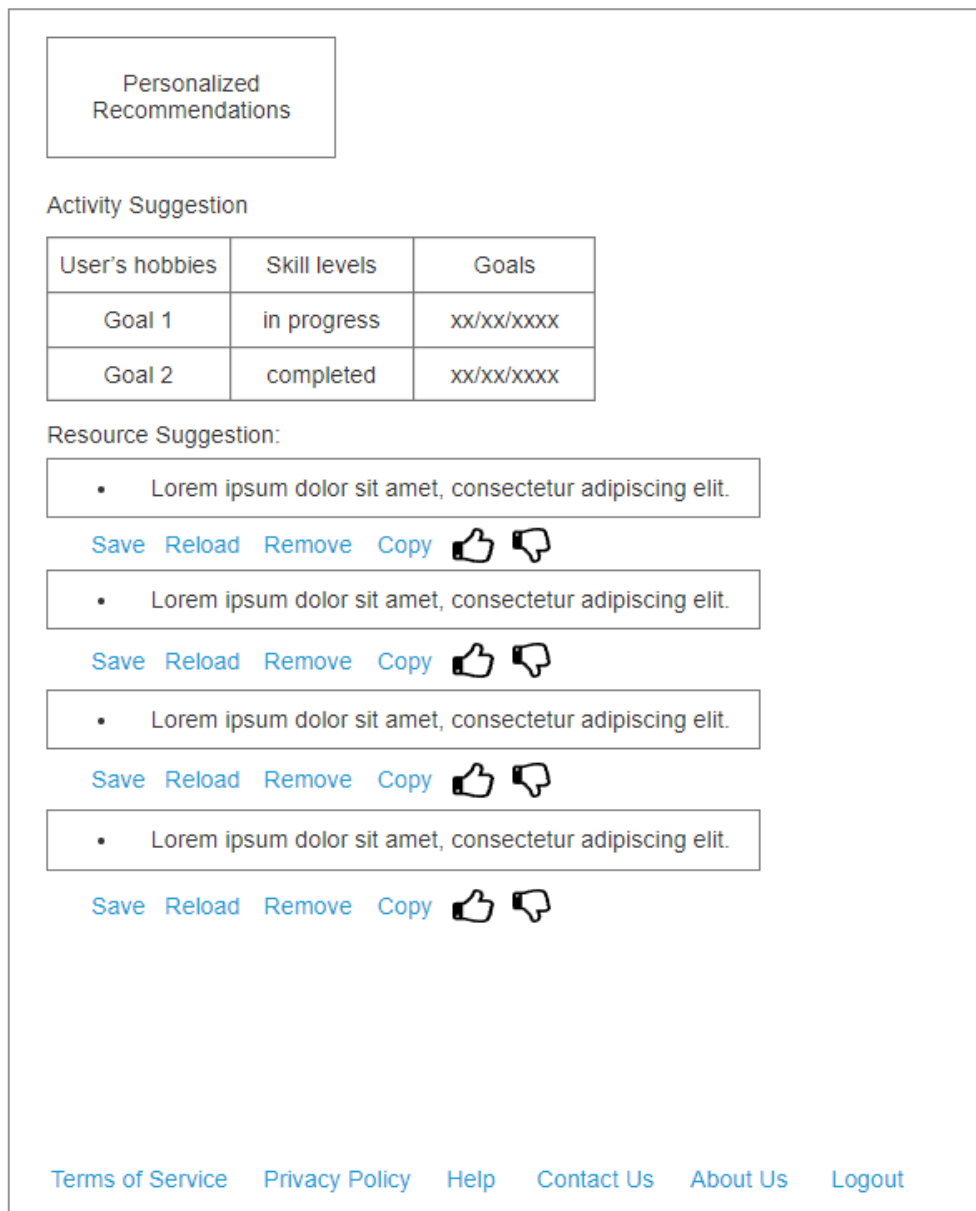


Figure 4.7.9 Personalized Recommendation Page

Resource Library

Search:

Advanced Filters ▼

Category List:

☐ Photography

☐ Coding

☐ Reading

☐ Video games

☐ Writing

☐ Music

☐ Sports

☐ Cooking

☐ Gardening

☐ Arts

☐ Crafts

☐ Running

Subcategories:

☐ Beginner

☐ Intermediate

☐ Advanced

Popular ▼

 Author:

Title:



Description:

Share

Bookmark

Comment:

[Terms of Service](#)

[Privacy Policy](#)

[Help](#)

[Contact Us](#)

[About Us](#)

[Logout](#)

Figure 4.7.10 Resource Library Page

Community

Post:

Cover:

Add Cover

Title:

Description:

Select Tag:

Reading

Post

Popular



Author:

Title:



Description:

Share   Bookmark

Comment:

[Terms of Service](#) [Privacy Policy](#) [Help](#) [Contact Us](#) [About Us](#) [Logout](#)

Figure 4.7.11 Community Support

System Administration

User Overview

User Statistics:

Total Users: xxxxx

User Growth Chart:



User Activity Summary:

User	Average time spent
aaa	
bbb	

User Search: name / email / user ID

User List:

User ID	Name	Email	Status	Last login	Action
					Activate ▼
					Activate ▼

Content Overview

Content Statistics:

Total Contents: xxxxx

Content Growth Chart:



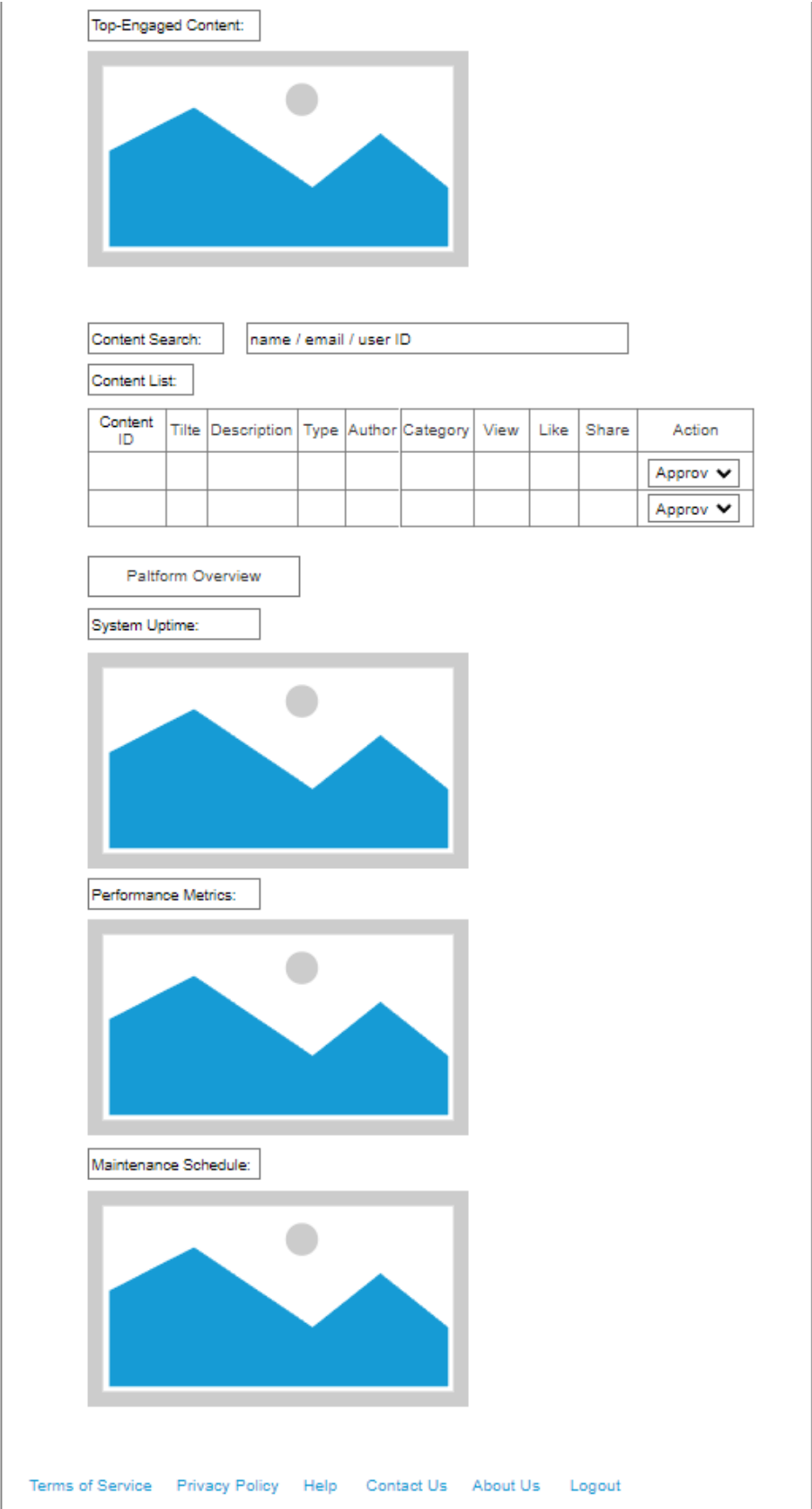


Figure 4.7.12 System Administration

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APPENDICES

Appendix A: Preview of the Questionnaire Questions

Questionnaire for AI-Enhanced Platform for Personalized Hobby Improvement

Introduction

Hello,

My name is Ricky Tang Ruey Horng, and I am a Year 3 student at Universiti Tunku Abdul Rahman (UTAR). As part of my Final Year Project, I am developing an AI-enhanced platform designed to help users improve their hobbies by providing personalized recommendations and resources. This project aims to bridge the gap in existing platforms by offering tailored guidance and insights to support users in their hobby pursuits.

To achieve this, I am conducting a survey to gather requirements and understand the needs and preferences of hobby enthusiasts. Your participation in this questionnaire will provide valuable insights that will help shape the development of the platform, ensuring it effectively addresses user needs and offers meaningful support.

Instructions

Please take a few minutes to answer the following questions. Your responses will be kept confidential and will be used solely for the purpose of this project. The questionnaire is divided into several sections, covering personal information, hobby-related experiences, and your thoughts on the use of artificial intelligence to enhance hobby improvement.

Your answers are **ONLY** being used for academic purposes and will be kept private.

We would like to thank you all in advance for your participation.

rickytangrh@utar.my [Switch account](#)

* Indicates required question

Email *

☒ Record rickytangrh@utar.my as the email to be included with my response

[Next](#) [Clear form](#)

Never submit passwords through Google Forms.

This form was created inside of Universiti Tunku Abdul Rahman. [Report Abuse](#)

Questionnaire for AI-Enhanced Platform for Personalized Hobby Improvement

rickytangrh@1utar.my [Switch account](#)



Your email will be recorded when you submit this form

Demographic Information:

What is your age?

- ☐ Under 18
- ☐ 18 - 25
- ☐ 25 - 40
- ☐ 40 - 60
- ☐ Older than 60

What is your gender?

- ☐ Male
- ☐ Female

What is your occupation?

- ☐ Student
- ☐ Employed full-time
- ☐ Employed part-time
- ☐ Self-employed
- ☐ Unemployed
- ☐ Retired

What is your highest level of education?

- ☐ Secondary School or below
- ☐ Diploma
- ☐ Bachelor's degree
- ☐ Master's degree
- ☐ Doctorate
- ☐ Other: _____

[Back](#)

[Next](#)

[Clear form](#)

Questionnaire for AI-Enhanced Platform for Personalized Hobby Improvement

[rickytangrh@1utar.my](#) [Switch account](#) 

Your email will be recorded when you submit this form

Hobby Information:

What hobbies do you actively pursue? (Select all that apply)

- ☐ Reading
- ☐ Writing
- ☐ Music
- ☐ Sports
- ☐ Playing video games
- ☐ Cooking
- ☐ Gardening
- ☐ Arts and crafts
- ☐ Other: _____

How long have you been engaged in these hobbies?

- ☐ Less than 6 months
- ☐ 6 months to 1 year
- ☐ 1-3 years
- ☐ 3-5 years
- ☐ More than 5 years

How often do you participate in your hobbies?

- ☐ Daily
- ☐ Several times a week
- ☐ Weekly
- ☐ Several times a month
- ☐ Monthly
- ☐ Rarely

What motivated you to start these hobbies? (Select all that apply)

- ☐ Personal interest
- ☐ Recommendations from friends or family
- ☐ To learn a new skill
- ☐ To relax and unwind
- ☐ Other: _____

[Back](#) [Next](#) [Clear form](#)

Questionnaire for AI-Enhanced Platform for Personalized Hobby Improvement

rickytangrh@1utar.my [Switch account](#)



Your email will be recorded when you submit this form

Motivation for Artificial Intelligence(AI) in Hobby Improvement

How important is it for you to receive personalized recommendations for your hobbies?

	1	2	3	4	5	
Not important	☐	☐	☐	☐	☐	Very important

What specific aspects of your hobbies would you like personalized recommendations for? (Select all that apply)

- ☐ Skill improvement
- ☐ New activities
- ☐ Troubleshooting
- ☐ Resources and content
- ☐ Other: _____

What difficulties have you encountered with current hobby management tools? (Select all that apply)

- ☐ Lack of personalization
- ☐ Poor user interface
- ☐ Limited resources
- ☐ Difficulty tracking progress
- ☐ Other: _____

How do you currently find resources for your hobbies? (Select all that apply)

- ☐ Online search
- ☐ Recommendations from friends or family
- ☐ Social media
- ☐ Dedicated hobby websites or apps
- ☐ Other: _____

What gaps do you notice in the available content for your hobbies? (Select all that apply)

- ☐ Outdated information
- ☐ Lack of variety
- ☐ Limited expert advice
- ☐ Inaccessibility
- ☐ Other: _____

How do you feel about using AI to provide suggestions for improving your hobbies?

	1	2	3	4	5	
Very negative	☐	☐	☐	☐	☐	Very positive

What would make AI-generated recommendations more useful for you? (Select all that apply)

- ☐ High accuracy
- ☐ Relevance to current skill level
- ☐ Variety in suggestions
- ☐ Ease of implementation
- ☐ Other: _____

How often do you follow recommendations for hobby improvement?

☐ Always

☐ Often

☐ Sometimes

☐ Rarely

☐ Never

What type of content do you find most helpful for improving your hobbies? (Select all that apply)

☐ Articles

☐ Videos

☐ Tutorials

☐ Online courses

☐ Community forums

☐ Other: _____

What features would you expect in a hobby management interface? (Select all that apply)

☐ Progress tracking

☐ Personalized recommendations

☐ Resource library

☐ Community support

☐ Goal setting

☐ Other: _____

How important is ease of use in managing your hobbies?

1 2 3 4 5

Not important ☐ ☐ ☐ ☐ ☐ Very important

What kind of visual elements (charts, graphs, etc.) would you find useful in tracking your hobby progress? (Select all that apply)

☐ Progress charts

☐ Skill improvement graphs

☐ Activity timelines

☐ Achievement badges

☐ Other: _____

☐ Send me a copy of my responses.

[Back](#) [Submit](#) [Clear form](#)

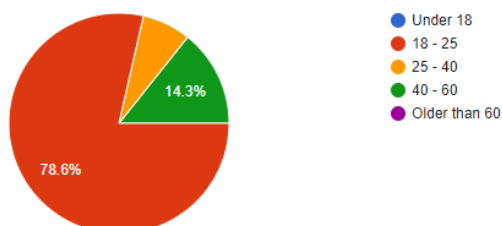
Appendix B: Response to the Questionnaire Questions

Demographic Information:

What is your age?

14 responses

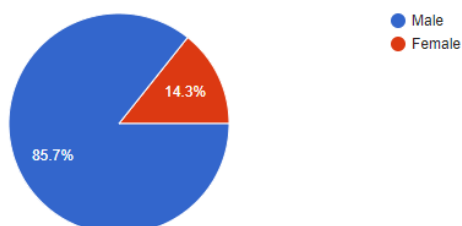
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What is your gender?

14 responses

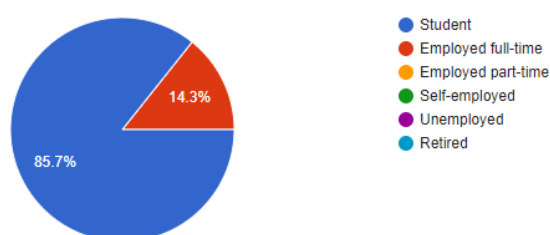
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What is your occupation?

14 responses

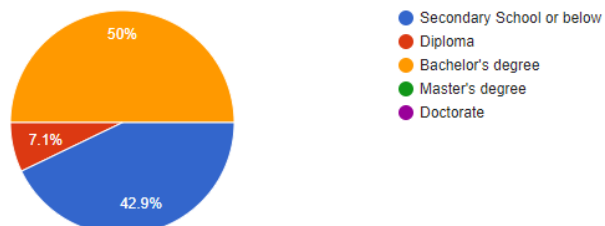
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What is your highest level of education?

14 responses

 Copy

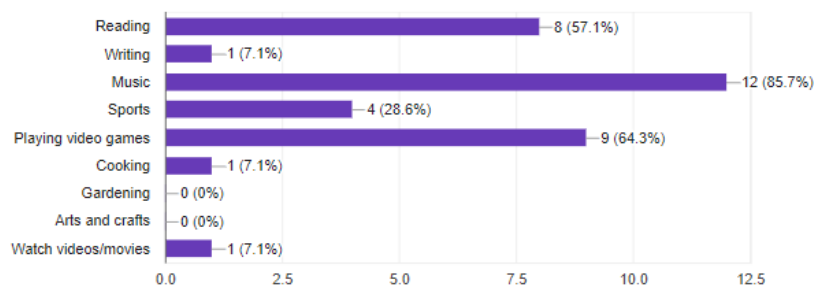


Hobby Information:

What hobbies do you actively pursue? (Select all that apply)

[Copy](#)

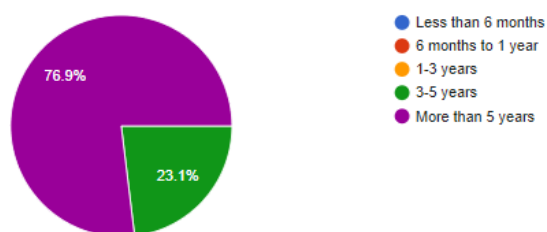
14 responses



How long have you been engaged in these hobbies?

[Copy](#)

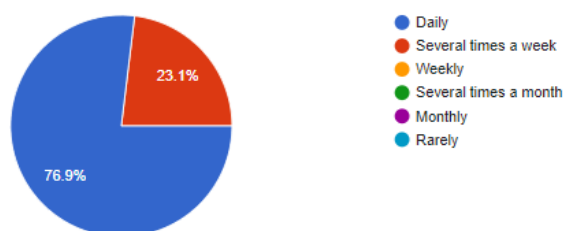
13 responses



How often do you participate in your hobbies?

[Copy](#)

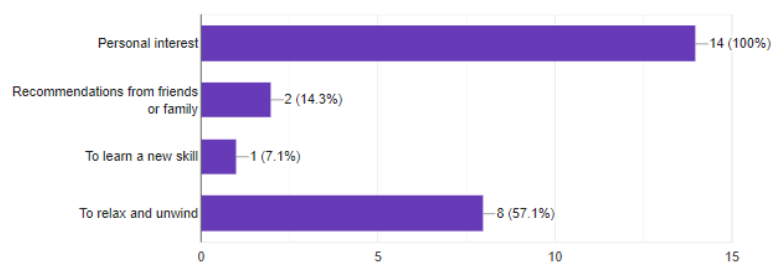
13 responses



What motivated you to start these hobbies? (Select all that apply)

[Copy](#)

14 responses

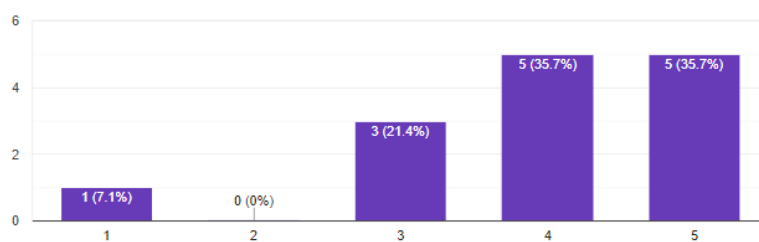


Motivation for Artificial Intelligence(AI) in Hobby Improvement

How important is it for you to receive personalized recommendations for your hobbies?

[Copy](#)

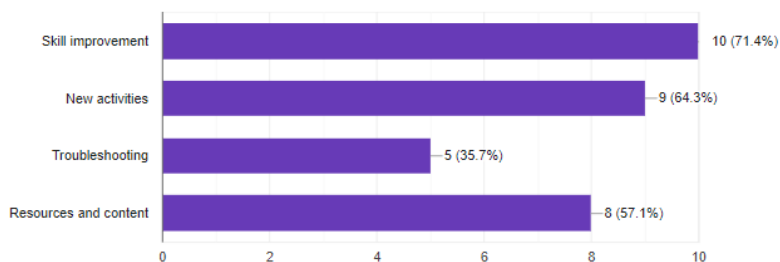
14 responses



What specific aspects of your hobbies would you like personalized recommendations for? (Select all that apply)

[Copy](#)

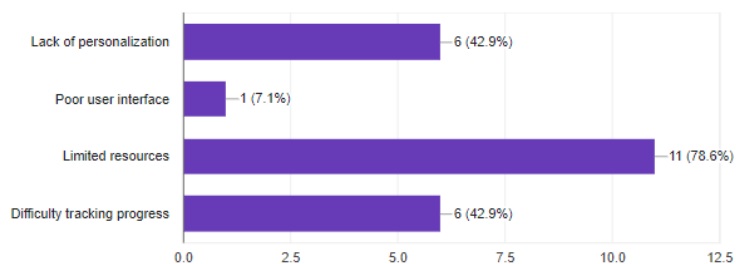
14 responses



What difficulties have you encountered with current hobby management tools? (Select all that apply)

[Copy](#)

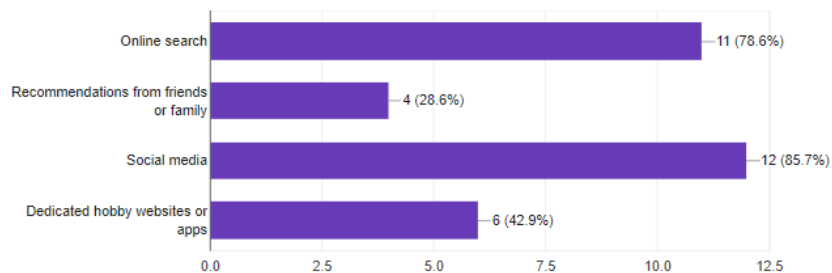
14 responses



How do you currently find resources for your hobbies? (Select all that apply)

[Copy](#)

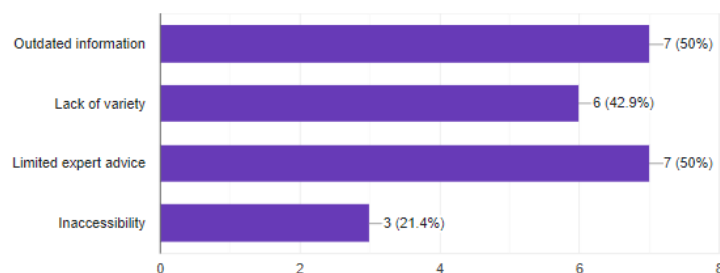
14 responses



What gaps do you notice in the available content for your hobbies? (Select all that apply)

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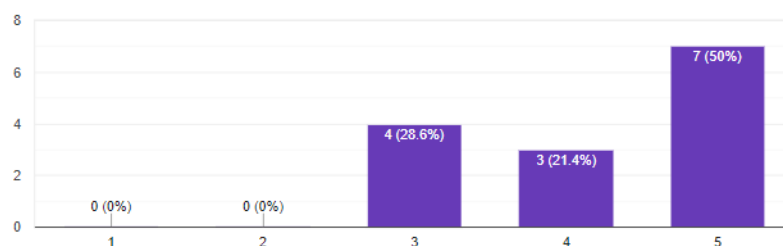
14 responses



How do you feel about using AI to provide suggestions for improving your hobbies?

[Copy](#)

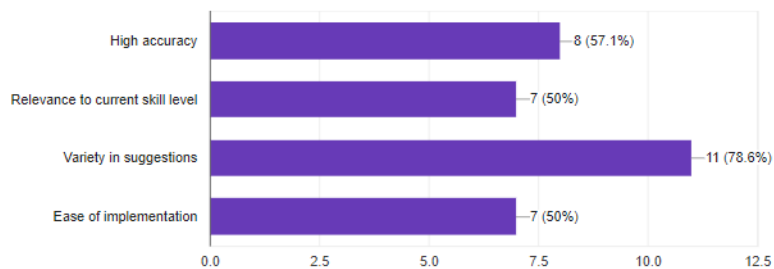
14 responses



What would make AI-generated recommendations more useful for you? (Select all that apply)

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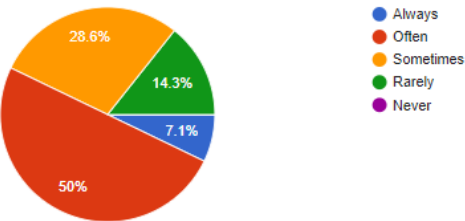
14 responses



How often do you follow recommendations for hobby improvement?

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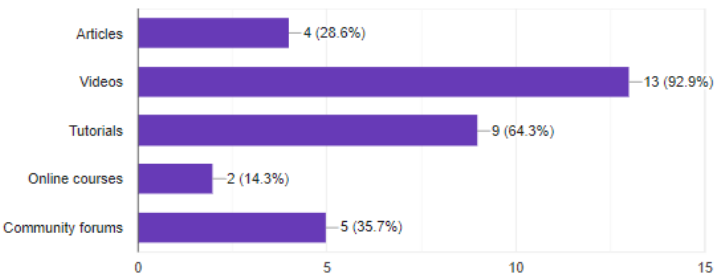
14 responses



What type of content do you find most helpful for improving your hobbies? (Select all that apply)

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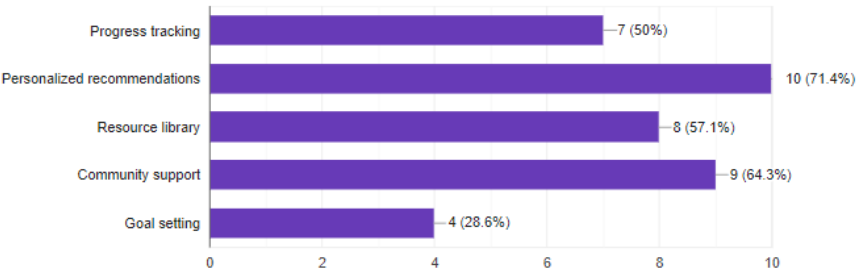
14 responses



What features would you expect in a hobby management interface? (Select all that apply)

 Copy

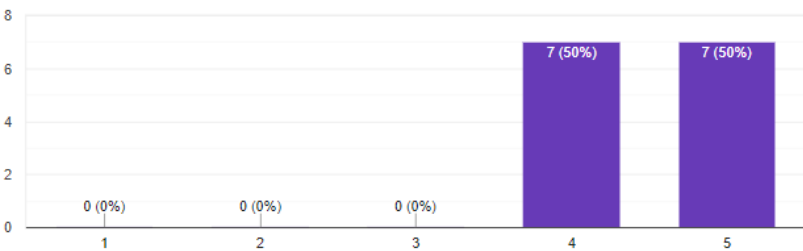
14 responses



How important is ease of use in managing your hobbies?

 Copy

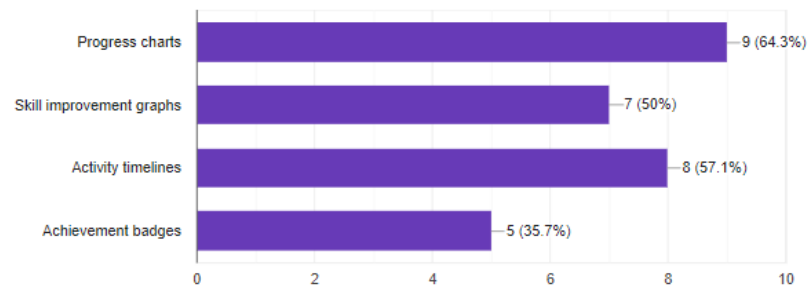
14 responses



What kind of visual elements (charts, graphs, etc.) would you find useful in tracking your hobby progress? (Select all that apply)

 Copy

14 responses



Appendix C: Overview of Gantt Chart

